

Enterprise Campus

Today's campus networks are almost all provisioned with Gigabit/10 Gigabit of bandwidth across the LAN, in which QoS might be seen as an unnecessary service to be considered because the need for queuing is minimal or almost not required as compared to the WAN and Internet edge, where queuing is a primary function to enhance the service quality. Although this statement is valid to some extent, the need for QoS is not only limited to perform queuing functions.

The unified marking and accurate traffic classification (as close to the source as possible) QoS can offer, when enabled across the campus LAN, also enables policing across the campus LAN to provide the flexibility and control to network operators to manage traffic usage based on different fields of traffic flows, such as ToS values. It can also be used as a protective mechanism in situations like DoS attacks, to mitigate their impact. Therefore, it is recommended that QoS be enabled across the campus LAN to maintain a seamless DS domain design in which classification and marking policies establish and impose trust boundaries, while policers help to protect against undesired flows at the access edge and across the LAN.

Enterprise Edge

Enterprise edge (WAN, extranet, or Internet) is the most common place that traffic flow aggregation occurs (where more or a larger number of traffic flows usually arrive from the LAN side and need to exit the enterprise edge, which is usually provisioned with lower capacity). For instance, the LAN side might be provisioned with Gigabit/10 Gigabit, whereas the WAN has only 10 Mbps of actual available bandwidth. Therefore, QoS is always one of the primary functions considered at the enterprise edge to achieve more efficient bandwidth optimization, especially for enterprises that are converging their voice, video, and data traffic. Logically, the enterprise edge represents the DS domain edge where mapping and profiling of traffic flows to align with the adjacent DS domain is required to maintain a seamless QoS design. For instance, [Figure 10-37](#) shows an example of a 12-class to 4-class QoS model mapping at the enterprise WAN edge router (CE) toward the SP edge (PE) to achieve end-to-end consistent QoS policies.

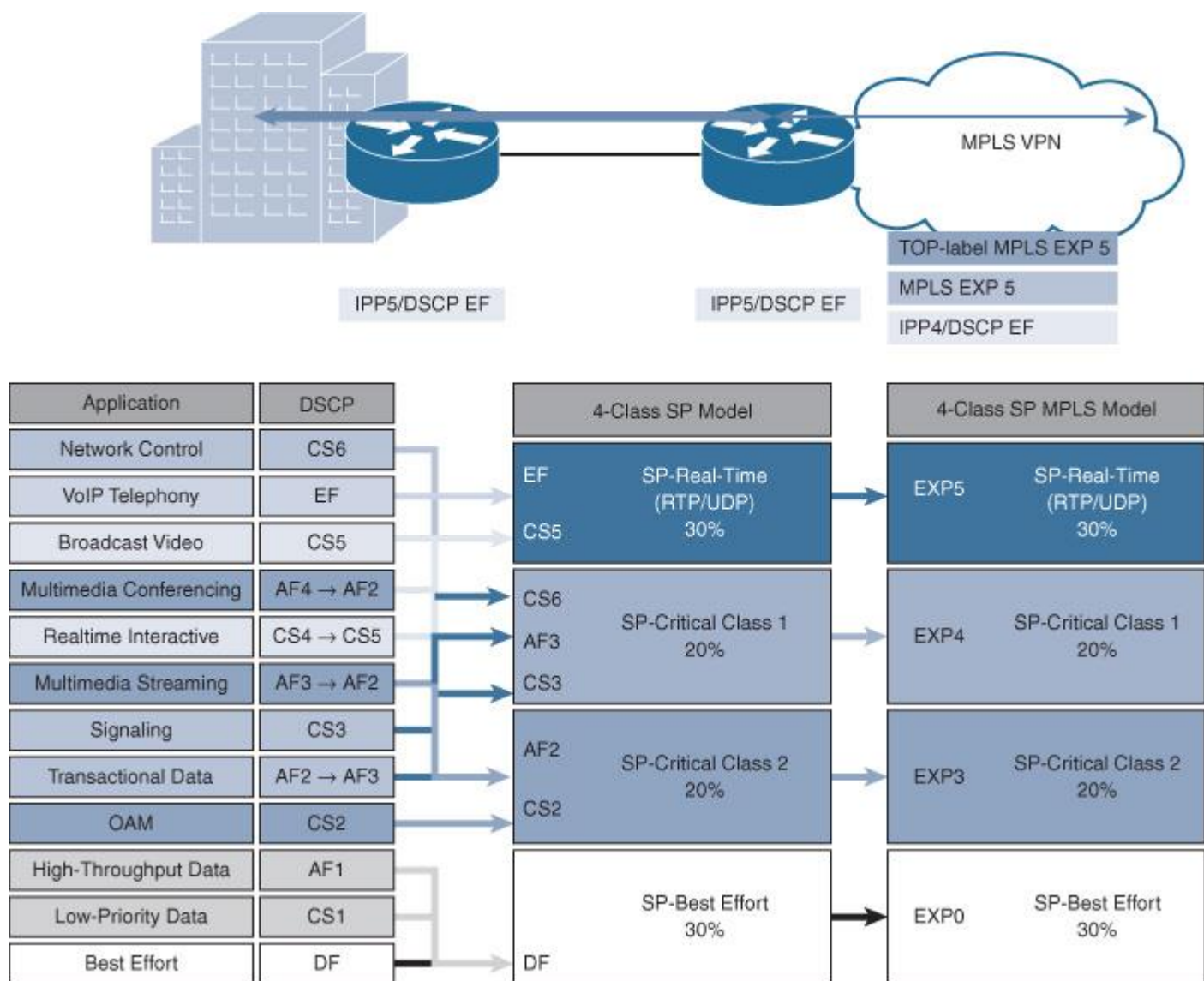


Figure 10-37 QoS Mapping: Enterprise WAN Edge

Note

The allocated bandwidth percentage in this figure is only for the purpose of the example and based on the best practices recommendations. However, these values technically must be allocated based on the SLA between the SP and the enterprise customer.

Note

It is common that SPs offer their CoS based on IP Precedence marking only. As a result, the enterprise marking based on DSCP and deployed with the 8- or 12-class model may encounter inconsistent end-to-end markings. For instance, if there is a video RTP stream sent out from one site and its packets are marked with DSCP value of “DSCP 34” or AF41 (in binary, 100010), it will convert to IPP 4 (in binary, 100). In turn, it will come back as DSCP 32 (binary, 100000) at the other remote site. Therefore, a re-marking is required in this case at the other side (receiving) in the ingress direction to maintain a unified QoS marking end to end. Also, the used “MPLS DiffServe Tunneling mode” and its impact on the original DSCP marking across the SP network is covered later in this chapter.

IP Tunneling QoS Design Considerations

As discussed earlier in this book, the attractive pricing drives many enterprises to consider deploying different VPN solutions as either alternative solutions to the private WAN transport or as redundant paths to the private WANs. In addition, some enterprises have to comply with their security policy that dictates considering a secure IP transport across the WAN/MAN, even if it is private, in which case the enterprise may consider IPsec, secure DMVPN, or GETVPN to protect the communication over the WAN or MAN transport.

One primary concern is the impact of increased packet overhead on the existing applications, because each VPN solution will add additional IP and ESP headers, as shown in [Figure 10-38](#). In addition, VPN brings a serious concern about bandwidth consumption. For instance, at Layer 3, a G.711 VoIP RTP stream needs 80 kbps. When this stream is sent over a GRE tunnel encrypted with IPsec, the total required bandwidth increases to ~112 kbps, which is an almost 40-percent bandwidth increase to transport the encrypted VoIP call. Therefore, bandwidth consumption with VPN is an essential point network designers need to consider, because it will usually impact the overall supported number of simultaneous voice calls or application sessions based on the actual available bandwidth.

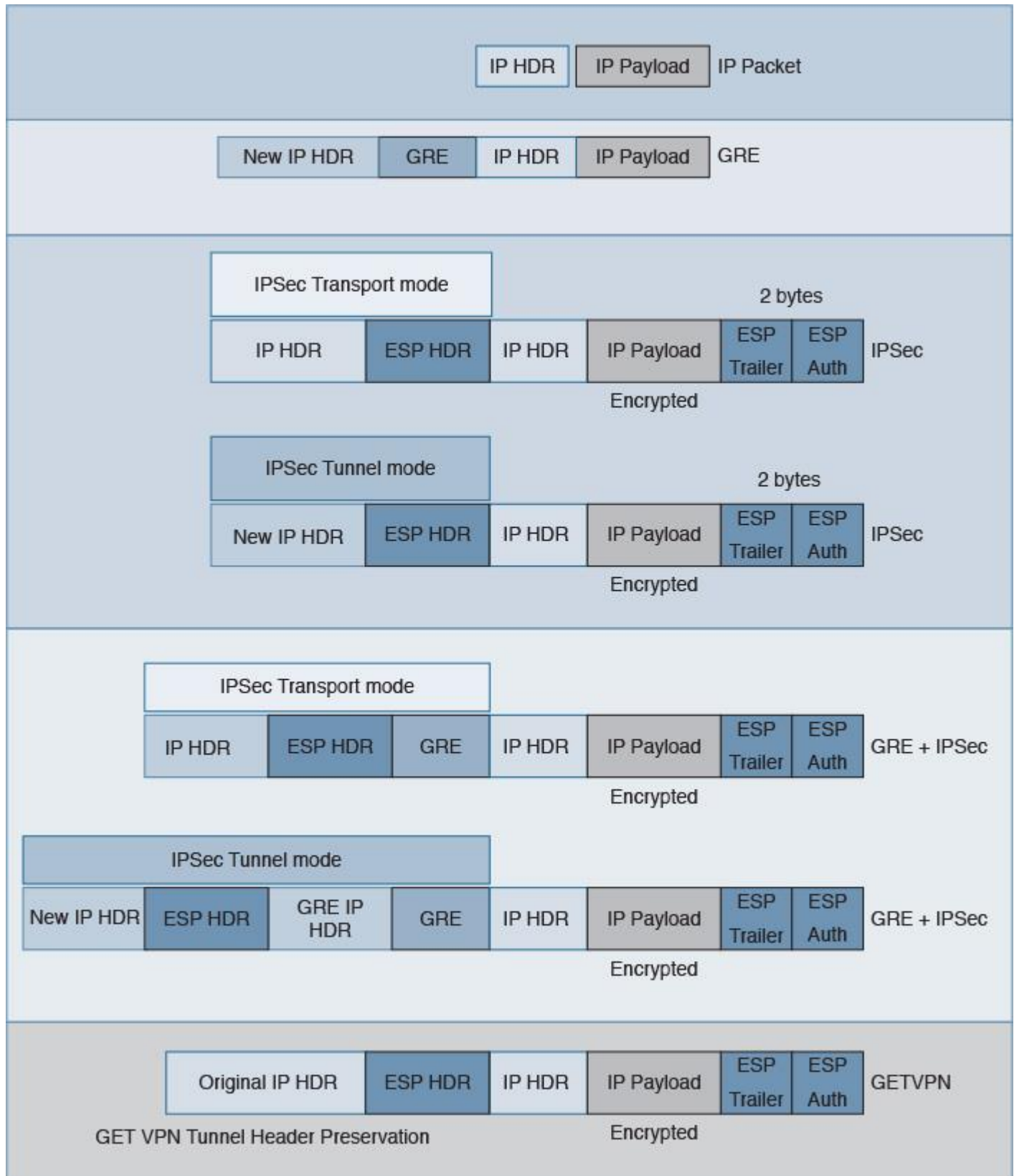


Figure 10-38 IP Packet Overhead with VPN Solutions

In addition, the following factors with regard to QoS design should also be taken into account by network designers when considering any VPN solution:

- Additional packet delay caused by the encryption and decryption functions.
- Pre-crypto queuing.
- The unforeseen impact of QoS packet reordering (prioritization) that may lead the IPsec anti-reply mechanism to drop legitimate but out-of-sequence packets.
- ToS value preservation.
- Maximum transmission unit (MTU) issues.
- Original flow information by default will be hidden for outbound QoS policies. (Cisco's QoS preclassify feature helps in this case.)
- Traffic flow with regard to the VPN logical topology versus the underlying IP transport topology. For instance, adding an overlay solution based on DMVPN phase 1 over an MPLS L3VPN WAN will transform the topology from any to any to hub and spoke for traffic sent over the DMVPN cloud.

Furthermore, extending enterprise converged services, such as voice and video over an overlay (VPN) solution, requires overlaying QoS techniques and tools over the enterprise VPNs to satisfy the different service levels for voice, video, and other data business applications.

In general, the following QoS toolset forms the foundation of QoS design with VPN solutions:

- **Hierarchical QoS:** As discussed earlier, HQoS helps in scenarios where QoS policies have to be applied on an interface with a sub-line rate. Similarly, with VPN and tunneling, the tunnel might use a fraction of the line rate of the interface, in which HQoS helps to optimize per tunnel QoS.
- **CBWFQ queuing:** This is required when different traffic treatment is required.
- **Admission control:** To provide per-child QoS class admission control.
- **WRED:** Helps to further optimize TCP-based applications.

With the typical point-to-point VPN, whether it is a classic IPsec or GRE (combined with IPsec or not), traffic flow is more deterministic because it always between two points or sites over the VPN tunnel. However, because the nature of this type of VPN is based on point-to-point tunneling, it has high level of operational complexities and scalability limitations in large deployments. (For example, network operators need to maintain manual QoS policies per tunnel at the central or hub site, which makes any simple update a nightmare for the operations teams.) In contrast, DMVPN introduces a challenge with regard to QoS design here because DMVPN works in a hub-and-spoke topology and offers the ability to route traffic directly between the spoke sites, in which traffic flow is directed between spokes as well (not to mention the scale of remote site DMVPN support, where tens of hundreds of spokes can be easily added and integrated with the DMVPN cloud). This adds operational complexity to managing and maintaining seamless QoS where the hub normally has a single big pipe with higher bandwidth that can easily congest the spokes. Adding QoS policy per spoke at the hub site is a nonscalable and inflexible option.

The per-tunnel QoS feature with DMVPN will promote a zero-touch hub design that supports NHRP groups to dynamically provision QoS policies on a per-spoke basis at the DMVPN hub node facilitated by the NHRP group feature. Remote sites (spokes) can be profiled based on the provisioned WAN or Internet bandwidth and then automatically assigned QoS policy per spoke at the hub mGRE tunnel during the spoke tunnel registration (as shown in [Figure 10-39](#)), in which it will shape the traffic from the hub (higher-bandwidth side) toward the spoke (lower-bandwidth side) to be always aligned with the spoke's maximum download rate.

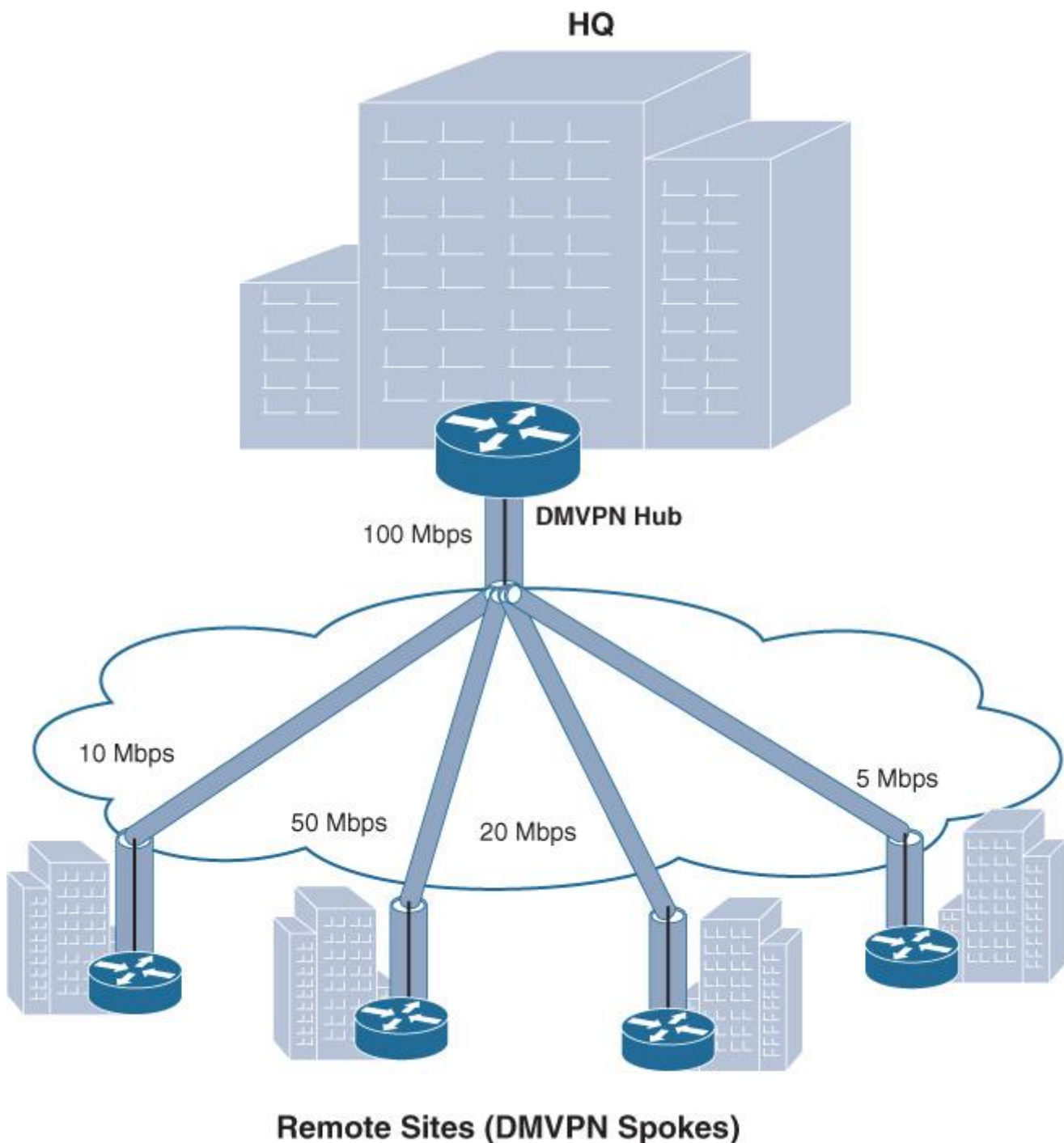


Figure 10-39 *DMVPN per Tunnel QoS*

Note

At the hub site, network operators still need to define a policy to shape the traffic of the interface to match the actual provisioned sub-line rate of bandwidth. Spokes, however, should follow the typical HQoS deployment per site, where the bandwidth of the Internet/WAN link must be shaped to the maximum upload provisioned capacity. This means direct spoke-to-spoke traffic streams will be controlled by the HQoS policies defined at the spokes level

One key issue network designers must be aware of when considering per-tunnel QoS for a DMVPN solution is that GRE, IPsec, and the L2 overhead must be considered when calculating the required bandwidth for QoS shaping and queuing, because these headers will be included as part of the packet size. This is because queuing and shaping, technically, is executed at the outbound physical interface of the DMVPN mGRE tunnel. GETVPN, in contrast, preserves the entire original IP packet header (source and destination IP addresses, TCP and UDP port numbers, ToS byte, and DF bit) simply because QoS is applied at each GETVPN group member because no tunnels are used. This makes GETVPN design and deployment simpler, and the same

standard WAN QoS design can be applied, with the exception that packet size will be increased and therefore must be considered as part of QoS bandwidth calculations.

Service Provider QoS Design

In modern SP networks, QoS is becoming a primary requirement because transporting Internet traffic, voice, video, cellular traffic, and MPLS VPN business traffic over a common IP backbone, each with different SLA expectations, will without any doubt introduce significant implications for both network design and operations. These implications are associated with many challenges, because each type of traffic has different bandwidth, jitter, and latency requirements. Failing to meet these requirements or an inability to satisfy the agreed SLAs will probably impact the SP business negatively, such as paying an expensive penalty, not to mention the impact on its reputation in the market. Therefore, QoS consideration in today's SPs is almost a must.

However, the consideration of QoS does not mean always to deploy QoS end to end (hop by hop). For instance, there might be an SP core network overprovisioned with high bandwidth where QoS will not practically add any value and network architects or designers may consider basic QoS only to protect important traffic flows in situations like double-failure scenarios within the core network. In fact, network designers, during the planning and design phase, can decide what QoS model, strategy, and tools are required to meet the SP business goals and priorities with regard to the offered services along with the utilized infrastructure, taking into consideration current traffic volume and the projected growth in the future.

All concepts, strategies, and methodologies covered earlier in this chapter are applicable. Therefore, this section covers only the considerations that are more specific to the SP style of network; all other concepts discussed earlier should be considered where applicable.

Traffic Marking Strategy

In SP types of networks and in large enterprise networks where the core is MPLS enabled while the edge of the network is non-MPLS such as the typical PE-CE connectivity model or an enterprise that has an MPLS core to interconnect multiple regional non-MPLS (native IP) networks (as shown in [Figure 10-40](#)), architects and network designers may need to consider different end-to-end traffic treatment with regard to marking based on the type of network, organization's objective, and application requirements. The following are the primary marking strategies (as defined in RFC 3270), also known as *MPLS DiffServe tunneling modes*. The selection of any mode or strategy will drive the overall QoS traffic marking and policy application. Therefore, they are the primary and most commonly used strategies to achieve a seamless end-to-end QoS with regard to the marking of IP traffic flowing across a single or multiple networks (DS domains), which in turn will influence the design and application of QoS queuing and admission control at the DS domains' boundaries.

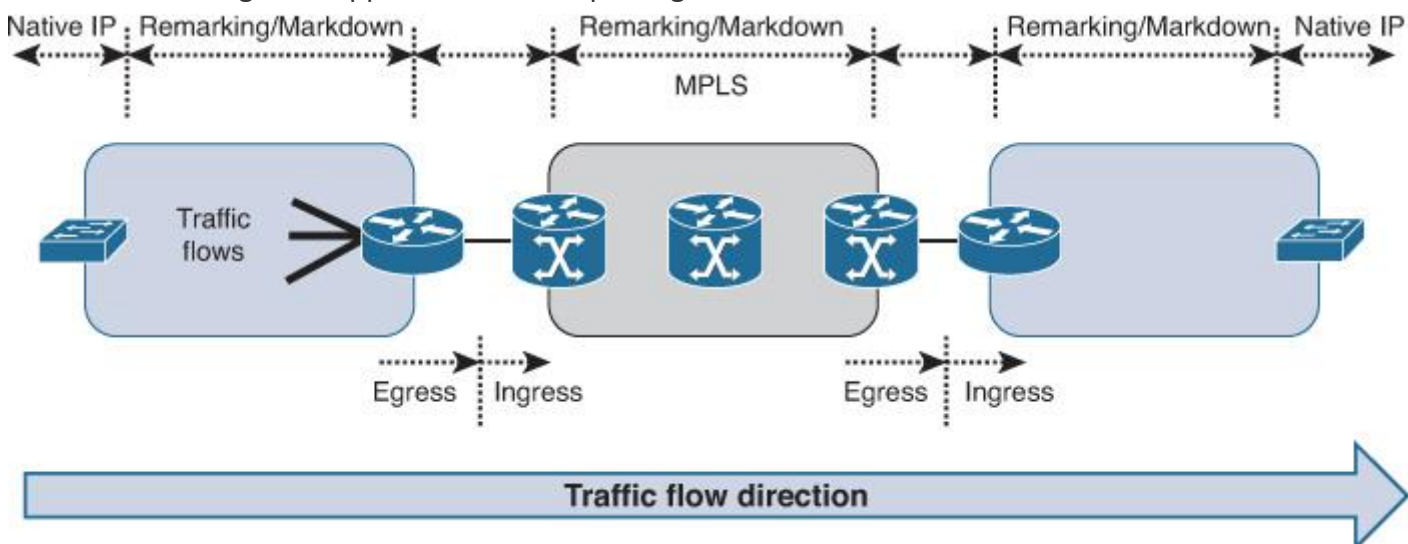


Figure 10-40 Multiple DS Domains with MPLS Enabled in the Core

Uniform Mode

With this mode, the first 3 bits of the IP ToS field (IP Precedence bits) will automatically be mapped to the MPLS EXP bits at the demarcation point between native IP and MPLS domains in the ingress direction and vice versa. This means that unified end-to-end marking values will be maintained across all the domains in a dynamic manner. In other words, both native IP and MPLS domains will be considered as a single DS domain.

If a re-marking is performed on a traffic flow that is out of contract within the MPLS domain, for example, this in turn will be reflected on the IP ToS value at the egress edge of the MPLS domain, as shown in [Figure 10-41](#). This is an easy and simple-to-enable solution. However, from the design point of view, this might not be practically acceptable in scenarios like an MPLS L3VPN provider offering SLA with differentiated services, where customers normally need their traffic marking to be maintained end to end. In contrast, this mode can be a suitable option for enterprises that deploy their own core MPLS and traffic marking values and re-marked values that should be maintained intact end to end.

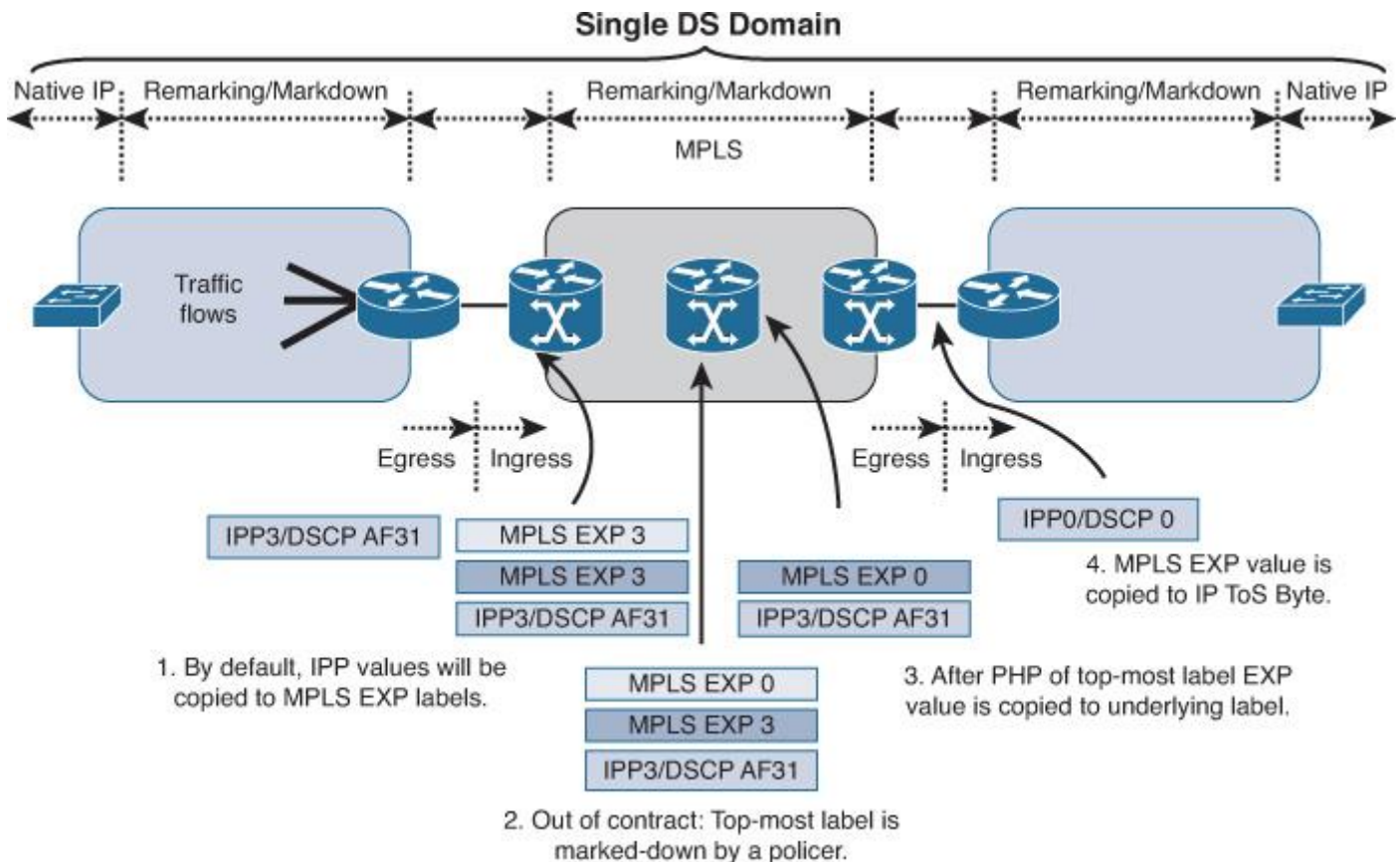


Figure 10-41 MPLS QoS: Uniform Mode

Pipe Modes

This mode consists of two flavors: the short and long modes. In general, both modes divide the QoS design with regard to the sample network in [Figure 10-41](#) (native IP with MPLS core) into two DS domains. A typical example here is an MPLS VPN provider and its customers. With this mode (pipe), any re-marking that happens to the MPLS EXP is limited to MPLS EXP, without impacting the underlying IP packet's ToS byte. As a result, the MPLS core, such as the MPLS VPN provider, will have the flexibility to re-mark (mark down) out-of-contract traffic and apply its own policies across the MPLS cloud without impacting the original IP marking set by the customer side. This, in turn, will offer MPLS VPN customers, at the same time, a seamless end-to-end IP marking to maintain their own end-to-end QoS policies in a more simplified way (where minimal or no need for traffic re-mark is required at the remote CEs in the ingress direction to align with the enterprise marking model). In addition, MPLS EXP is not mapped automatically from the IPP and is required to be defined explicitly on the ingress direction of the MPLS DS domain to align with its QoS policies or the agreed SLA.

The short and long pipe modes, however, differ in the way they define the demarcation boundary between MPLS EXP marking and IP IPP/DSCP marking at the egress edge (PE) of the MPLS DS domain. With the short pipe mode, the egress QoS policies at the PE node facing the CE are based on IPP or DSCP values, as illustrated in Figure 10-42.

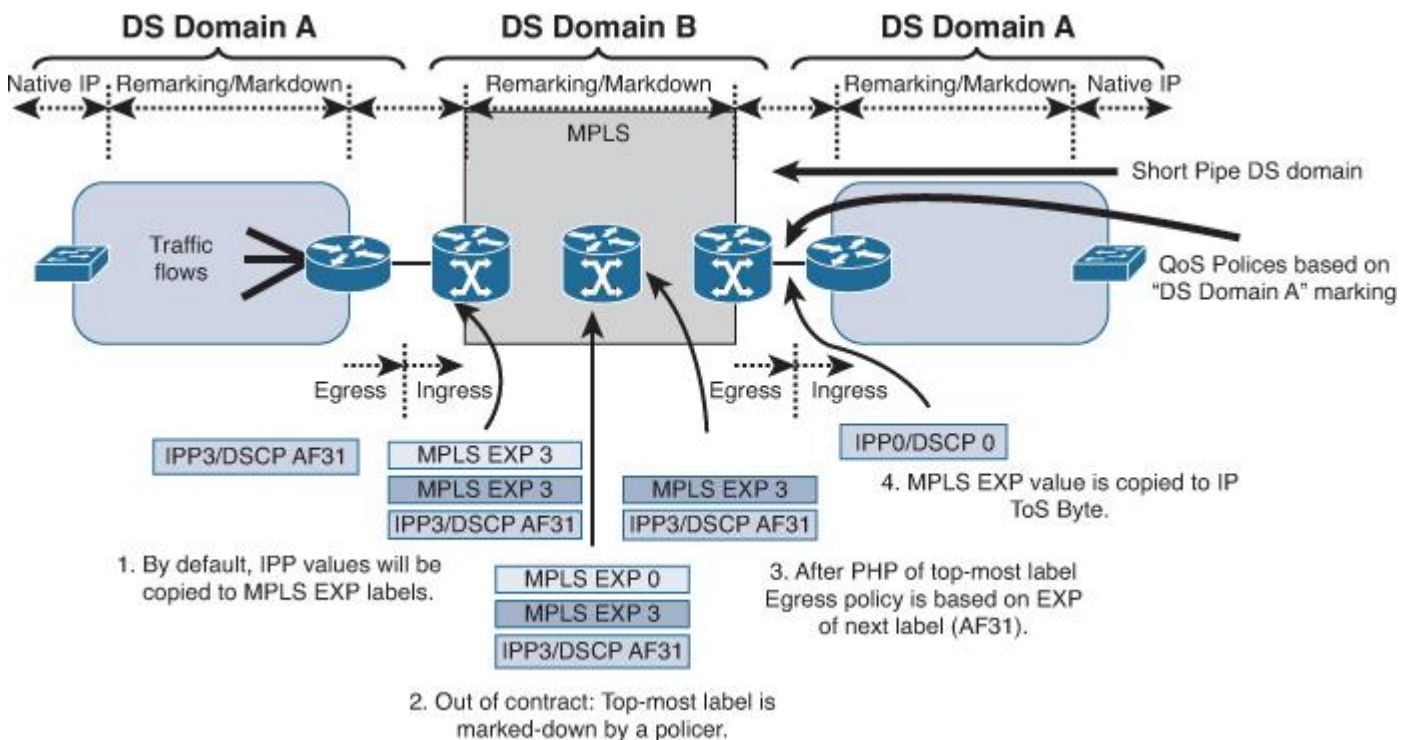


Figure 10-42 MPLS QoS: Short Pipe Mode

Note

The “long pipe mode” is sometimes referred to as “pipe mode.”

In contrast, the long pipe mode is based on the MPLS DS domain’s explicit markings and re-markings of EXP values, as shown in Figure 10-43.

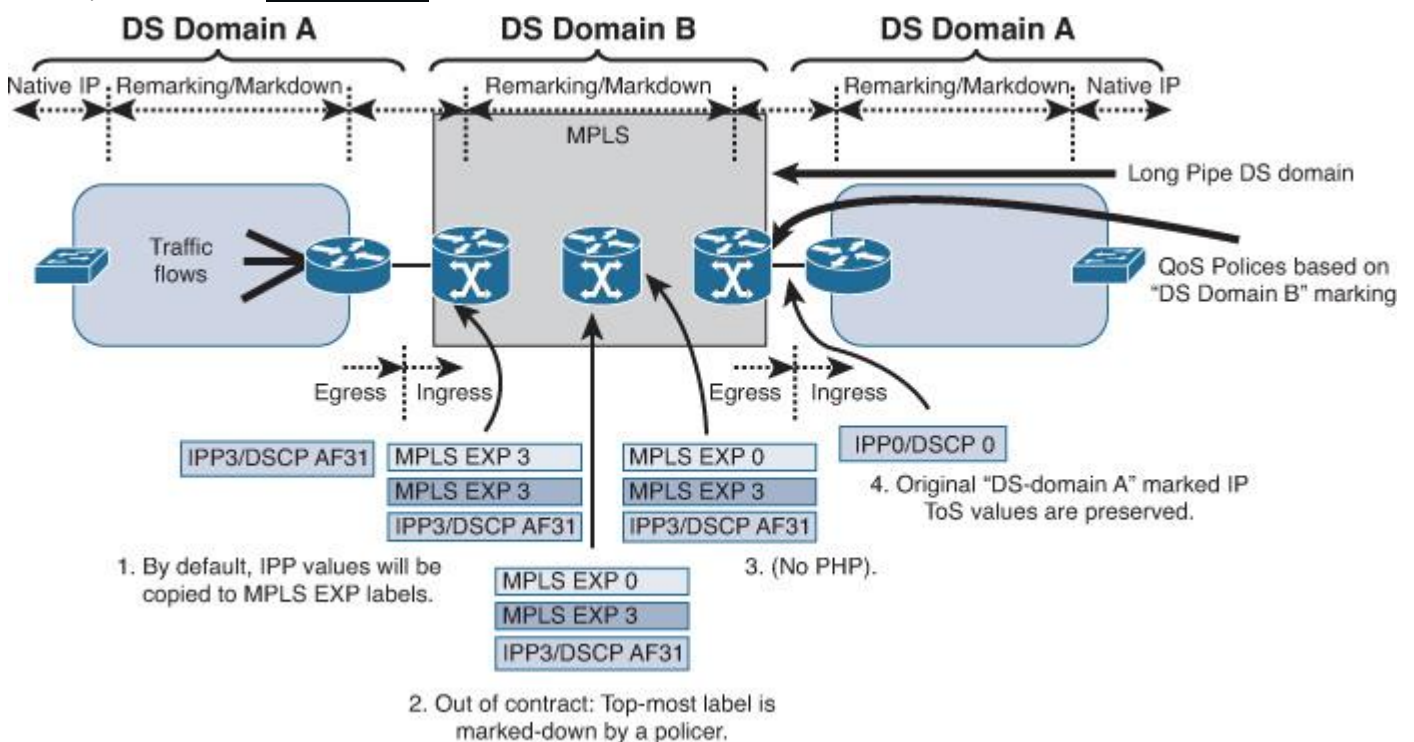


Figure 10-43 MPLS QoS: Long Pipe Mode

From a design point of view, the short pipe offers more desirable end-to-end QoS policies that align with the customer's (IP DS domain) if the scenario is a typical MPLS VPN. However, the short pipe mode may not be the right option if the design is a large-scale enterprise organization with MPLS core and the design goal is to treat out-of-profile or -contract traffic flows based on their re-marked EXP values when they exit the MPLS DS domain. (For example, a traffic flow marked as EXP 5, then re-marked to EXP0, should be treated as EXP0 by the QoS policy at the MPLS egress edge.) In other words, no one option is better than others; it all depends on the design requirements and how to align the QoS policies with design and strategy to offer a seamless scalable and flexible design.

Note

Managed CE is one of the common services operators (such as MPLS VPN providers) offer to their customers. In offering this service, the MPLS VPN providers are responsible for configuring and operating the CE nodes. In this type of scenario, some operators prefer to reduce some of the QoS complexity, such as marking and re-marking from their PE nodes and moving it to the managed CE nodes, where the operator usually can define the QoS policies with the agreed bandwidth allocation and marking. With regard to marking, it is always preferred to keep it as simple and consistent end to end as possible. Therefore, MPLS EXP marking is the preferred marking at those managed CEs. However, the challenge here is that the majority of the PE-CE connectivity is based on native IP (non-MPLS). At the same time, the operator must not impact any IP DiffServ markings set by the customer. Pipe mode MPLS DiffServ tunneling, with an Explicit Null LSP, can resolve this challenge (RFC 3032), where a label is assigned for customer traffic going toward the PE. This label is used only to preserve the MPLS VPN operator's MPLS EXP markings across the CE-to-PE link; the original customer IP marking is kept intact.

DiffServ MPLS-TE (DS-TE)

As discussed earlier in [Chapter 5](#), MPLS-TE introduces intelligence to the headend network node (source) to forward traffic based on multiple predefined criteria such as available bandwidth, link affinity attributes (like low-latency links), and SRLG considerations (a behavior known as *constraint-based routing*). DiffServ MPLS-TE (DS-TE) extends MPLS-TE to offer advanced class-based constraint routing of *guaranteed* traffic, in which MPLS-TE allows network operators to define two separate pools per link/path. Typically, the global pool reflects the overall available bandwidth over a given link (DS-TE), and the subpool reflects a subset of the available bandwidth for certain class such as EF (LLQ) class. It is important for the network designer to distinguish here between the standard DiffServ behavior discussed earlier in this section (such as CBWFQ and LLQ) and DS-TE, because DS-TE here can provide the control plane capability to book bandwidth at queue level (such as LLQ). Also, ideally it should align with the configured DiffServ LLQ bandwidth that provides the forwarding plane in this case.

In other words, DiffServ and DS-TE can complement each other (ideally). However, they are neither identical mechanisms, nor must they be enabled together. With DS-TE, network designers can achieve very advanced DiffServ-aware traffic engineering. For example, besides the standard ability of MPLS-TE to select a link to route traffic through based on certain criteria such as available bandwidth, DS-TE allows network designers to guarantee that the selected path can offer the available bandwidth at a class level, such as voice class. It also offers the ability to reroute traffic if a downstream node is experiencing congestion or has a limited available bandwidth to satisfy this class, which is impossible with the typical DiffServ to achieve because it is a per-hop mechanism (reactive local mechanism).

In addition DS-TE provides the ability to enable preemption priorities among DS-TE classes, where classes with higher priority can take precedence over those with lower priority during LSP signaling for the primary or backup MPLS-TE tunnel when there is not enough bandwidth available (tunnel preemption). Therefore, with MPLS DS-TE, SPs and large enterprises can offer end-to-end advanced QoS capabilities based on signaled rather than provisioned QoS to meet strict SLA requirements.

Note

With regard to MPLS DS-TE bandwidth constraints, there are two primary models: maximum allocation model (MAM) and Russian dolls model (RDM), as defined by IETF. The network designer may consider both models at the same time

at different places across the network. However, this approach will increase design and operational complexity and will reduce the end-to-end level of consistency and reliability of the solution.

Setting up the MPLS-TE LSPs (tunnels) is half the job because traffic will not ride the TE tunnel automatically. There are multiple mechanisms to route traffic inside the TE tunnel, such as autoroute, forwarding adjacency (FA), static route, and policy-based routing. As covered in [Chapter 5](#), each has its strengths and weaknesses. Technically, all these mechanisms are valid options to be used to route traffic down the MPLS DS-TE tunnels. However, the class-based tunnel selection (CBTS) mechanism offers a more flexible, scalable, and optimized TE solution when it is combined with MPLS DS-TE because it requires minimum configuration to dynamically route traffic over the desired DS-TE tunnels that takes into consideration the EXP marking of the traffic flows. In addition, CBTS offers the ability to support multiple TE tunnels of a given tunnel bundle to exit the headend node via different interfaces based on the assigned CoS (MPLS EXP value) to take different paths. For instance, traffic marked with EXP 5 can be sent out an interface that crosses a path with lower latency than others, as shown in [Figure 10-44](#).

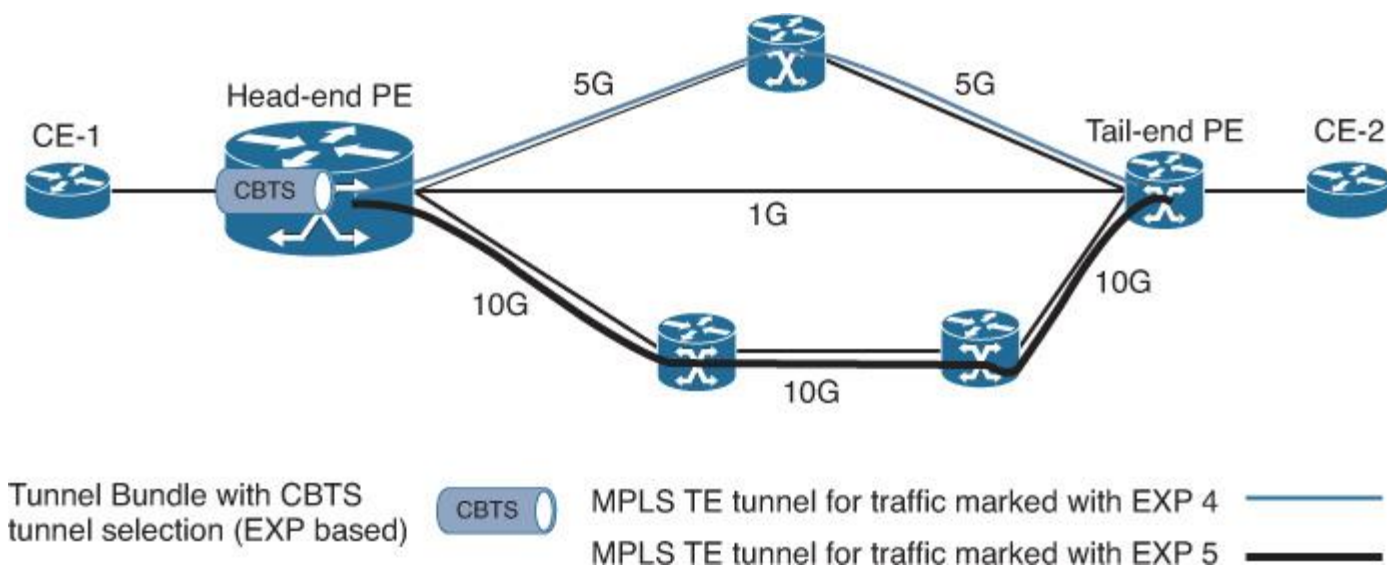


Figure 10-44 DS-TE CBTS Based

Note

Tunnel bundle refers to a group of normal TE or DS-TE tunnels that are sourced and destined from the same headend node to the same tail-end node. Typically, each can carry different MPLS EXP value.

Furthermore, one MPLS-TE primary design consideration discussed in [Chapter 5](#) is whether to set up the TE tunnel PE to PE or provider (P) to P, which depends on the targeted environment and the desired goal to achieve. Setting up a full mesh of PE-to-PE MPLS-TE tunnels can be a complex solution to plan, design, and operate in large-scale networks. Nonetheless, to offer a true end-to-end service differentiation with regard to DS-TE, it is desirable for the coverage of the TE tunnel and path optimization to be end to end. However, this does not mean every single TE tunnel has to be PE to PE. Instead, network designers can selectively decide which TE tunnels (such as DS-TE enabled only) should protect certain type of traffic flows to be set up PE to PE (outer-mesh based).

The critical point with regard to QoS design across MPLS-based networks (regardless of whether it is an SP or large enterprise network) is which design direction should be used, such as best effort, end-to-end DiffServ QoS, DS-TE based, or a combination. Obviously, no one can answer this question other than the designer who is aware of the different design and application requirements and the underlying network architecture. However, the following foundational questions help network designers to direct the design strategy toward the right path during the planning phase:

■ Is the targeted environment an SP or enterprise network? (Commonly, but not always, SP networks tend to have higher expectations in terms of traffic volume and network coverage and different SLA requirements.)

- Is the network bandwidth overprovisioned? (If yes, you may enable basic DiffServ QoS model.)
- Are there any sensitive applications that require strict SLA such as VoIP, video or any other mission/business-critical applications? If yes, if you fail to meet the end-to-end SLA for this traffic, will any penalty accrue, or will there be a business-critical impact? (Here you may decide to consider DS-TE [perhaps with MPLS-TE FRR] combined with a DiffServ model, if failing to meet the SLA has direct cost to or impact on the business.)
- Is MPLS-TE already enabled across the network? (If yes, this can simplify enabling DS-TE if required.)
- Does any inter-AS link require QoS considerations? (MPLS-TE over an inter-AS link can be a complex option to consider.)
- Is there any SRLG across the core?(If yes, MPLS-TE/DS-TE can overcome this design concern if an alternate path exists.)

As covered in [Chapter 1](#), design simplicity and avoiding gold plating are vital principles to be considered in every design. Therefore, keep in mind that the more protocols and features there are enabled across the network, the more complex the network is to design and operate.

Further Reading

“Enterprise Medianet Quality of Service Design 4.0—Overview,” <http://www.cisco.com>

“DiffServ—The Scalable End-to-End QoS Model,” <http://www.cisco.com>

“IPsec VPN QoS Design,” <http://www.cisco.com>

RFC 4124, *Protocol Extensions for Support of Diffserv-Aware MPLS Traffic Engineering*, <http://www.rfc-base.org>

RFC 3644, *Policy Quality of Service (QoS) Information Model*, <https://tools.ietf.org>

Network Security Design

Today’s converged networks carry much business-critical information across the network (whether it is voice, video, or data), which makes it extremely important for that information to be secured and protected for two primary reasons. The first reason is for information security and privacy purposes. The second reason is to maintain business continuity at the desired level, such as protecting against distributed DoS (DDoS) attacks, regardless of whether this protection is within the internal network or between the different sites over an external network. Therefore, the design of a network infrastructure and network security must not be performed in isolation.

In other words, the holistic design approach discussed earlier in this book is vital when it comes to network security design considerations. No network should be designed independently from its security requirements. A successful network design must facilitate the application and integration of the security requirements by following the top-down approach, starting from the business goals and needs, to compliance with the organization’s security policy standards, to a detailed design and integration of the various network technologies and components.

This section covers the different security aspects that network designers need to consider when planning, designing, or optimizing IP networks to achieve a self-healing, secure network infrastructures that can mitigate and operate with an acceptable level of performance during security breaches or network attack situations and that can offer the required level of information security across the network transports.

Note

Although the CCDE exam is not a security-focused exam, infrastructure security and the integration of security principles and methodologies with the network design are important aspects that every network designer must take into account to produce a successful design. In addition, one common challenge when optimizing an existing design is that network and security components are integrated in a way that either does not scale or does not provide the desired level of performance. Nonetheless, network designers are expected to identify the limitations in the design and optimize it in a way that meets the business and functional requirements. In other words, although the CCDE

candidate is not expected to produce deep-level security design, the candidate should know how to produce a secure network design by integrating multiple security principles in a way that does not break the network design goals while at the same time not compromising security requirements. Therefore, this section only covers network security design principles, approaches, and considerations that can impact the network design and the overall network design objective. It focuses mainly on how to achieve an integrated secure network design.

Network Security Design Fundamentals

This section discusses the primary design principles and considerations that network designers must take into account when designing or validating an existing network design from a network security point of view.

Top-Down Design

As discussed earlier in this book, to achieve a successful business-driven design, network architect and designers must always consider this approach to build the foundation or the roadmap of the design. With regard to security, in particular network security, network designers should consider the following questions when designing a new network design or when evaluating an existing network design to help them draw the high-level picture of the design direction with regard to the security aspects:

- What are the business objectives and goals?
- What is the targeted industry (manufacturing, financial services, retail)? (This helps to identify industry-specific standards.)
- Is there any security policy and standards the design must comply with? (This is specific to the identified industry, such as Payment Card Industry Data Security Standard [PCI DSS] compliance.)
- What is a must and what is desirable to be protected (priorities)?
- What is the impact of a failure to protect the intended systems/networks on the business (such as cost or reputation)?
- Will security services or appliances break any mission-critical applications? If yes, how can this risk be avoided or mitigated?
- Are there any capital expenditure (capex) or operational expenditure (opex) concerns with regard to network security?

Security Policy Considerations

To secure any system or network, there must be predefined goals to achieve and specifications to comply with to ensure that the outcomes are measurable and always meet the organization standards. Therefore, to achieve this, organizations almost always develop security standards, policies, and specifications that all collectively aim to achieve the desired goal with regard to information security. This is what is commonly known as a *security policy*. A security policy is a formal statement of the rules by which people who are given access to an organization's technology and information assets must abide. It should also specify the mechanisms through which these requirements can be met and audited for compliance with the policy [86].

As a result, a good understanding of the organization's security policy and its standards is a crucial prerequisite before starting any network design or optimizing an existing design. This understanding ensures that any suggested solution will comply with the security policy standards and expectations of the business with regard to information security.

For instance, you may suggest redesigning an existing 1G dark fiber (owned by the organization) to a virtual leased line (VLL) solution (L2VPN based) that offers the same quality with lower cost. However, the security policy may dictate that any traffic traversing any network that is not owned by the organization must be encrypted. By taking this point into consideration, the network architect or designer here can add IPsec or MACsec to the proposed solution to provide an encrypted VLL, to ensure the suggested design supports and complies with the organization's security policy standards.

Holistic Approach Considerations

The integration of network infrastructure and network security (including security components such as firewalls and configurations such as infrastructure access control lists ACL [iALCs]) can be seen as a double-edged sword. This is because on the one hand, security will offer privacy, control, and stability to the network (for example, protect against DDoS attacks and unauthorized access). On the other hand, if both the network infrastructure and the security components are designed in isolation (siloe approach), then when they integrate together at some point, there will be a mix of the following issues:

- Complex integration
- Traffic drop
- Reduced performance
- Redesign or major design changes

For instance, in [Figure 10-45](#), the network infrastructure of the Internet edge was designed to provide Internet access for the enterprise. Based on this, the network designer considered Enhanced Interior Gateway Routing Protocol (EIGRP) as the internal dynamic routing protocol and external Border Gateway Protocol (eBGP) to provide the edge routing to provide end-to-end connectivity.

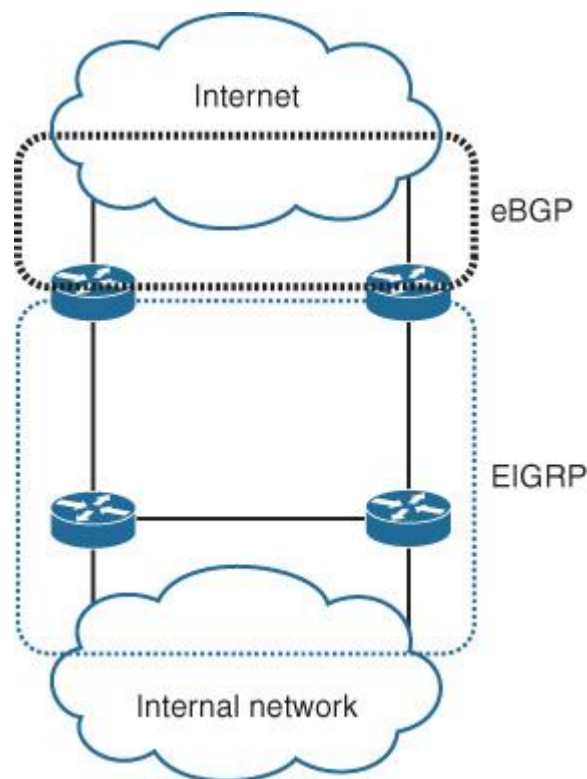


Figure 10-45 Network Design Before Inserting Security Devices

However, after this design was reviewed by the security team, they suggested that a pair of firewalls in routed mode has to be introduced into this design, and remote-access VPN sessions have to be terminated on these firewalls to comply with the enterprise security standards, as shown in [Figure 10-46](#). In addition, the added firewalls support only static routing.

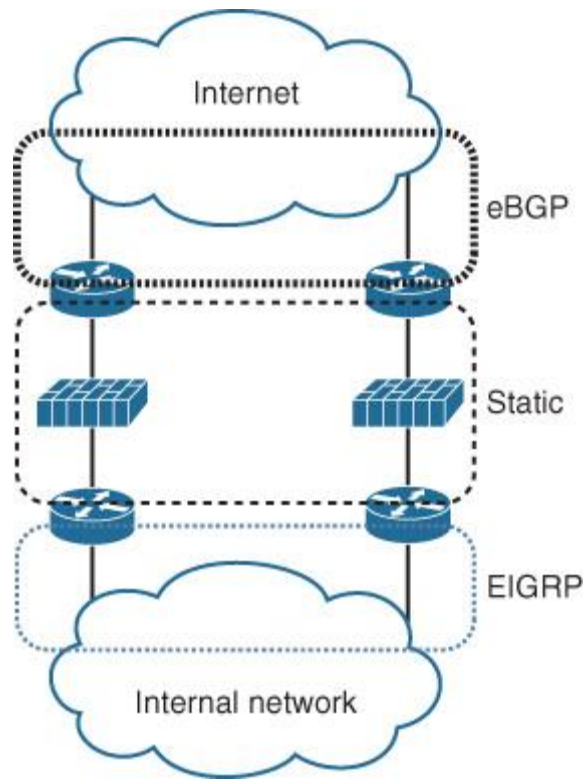


Figure 10-46 Network Design After Inserting Security Devices

Although it seems to be a simple change to introduce the firewalls into the original design, this simple change requires review of the entire design (IGP, BGP, inbound and outbound traffic flow, and policies to avoid asymmetrical routing, NAT setup on the edge routers to allow users' remote-access VPN sessions to terminate at the firewalls, in addition to whether these firewalls have any impact on the traffic flows of mission-critical applications passing through it). Therefore, it is important to adopt a holistic approach to promote an optimal, integrated, and secure network design that is flexible enough to achieve the intended goals with minimal operational and design changes.

Divide-and-Conquer Approach

To simplify the overall security design in a more controlled manner, a structured modular approach should be used by splitting the network into multiple domains. This approach helps to introduce *chokepoints* between the different domains to enforce structured control between these domains. Each domain should be given a level of trust based on the security requirements to facilitate controlling who and what is allowed to pass between the different domains by using various mechanisms such as packet filtering with iACLs or specialized security appliances (physical or virtual) such as a firewalls and intrusion prevention systems (IPS) to be positioned at each chokepoint (domain boundary point) between the different security domains.

Furthermore, security requirements sometimes dictate that within each security domain there must be subdomains where each should have its own specific security policies and access restrictions among other subdomains. In this case, network designers can introduce what is commonly known as *zones*. For example, a data center can be placed under a single security domain. Normally, inside the data center various services and applications require different levels of security enforcement between them, like web servers that can be placed in their own zone, database servers in different zones, and so on. With this approach, network designers can facilitate the security control to a large extent and optimize its manageability, as shown in [Figure 10-47](#).

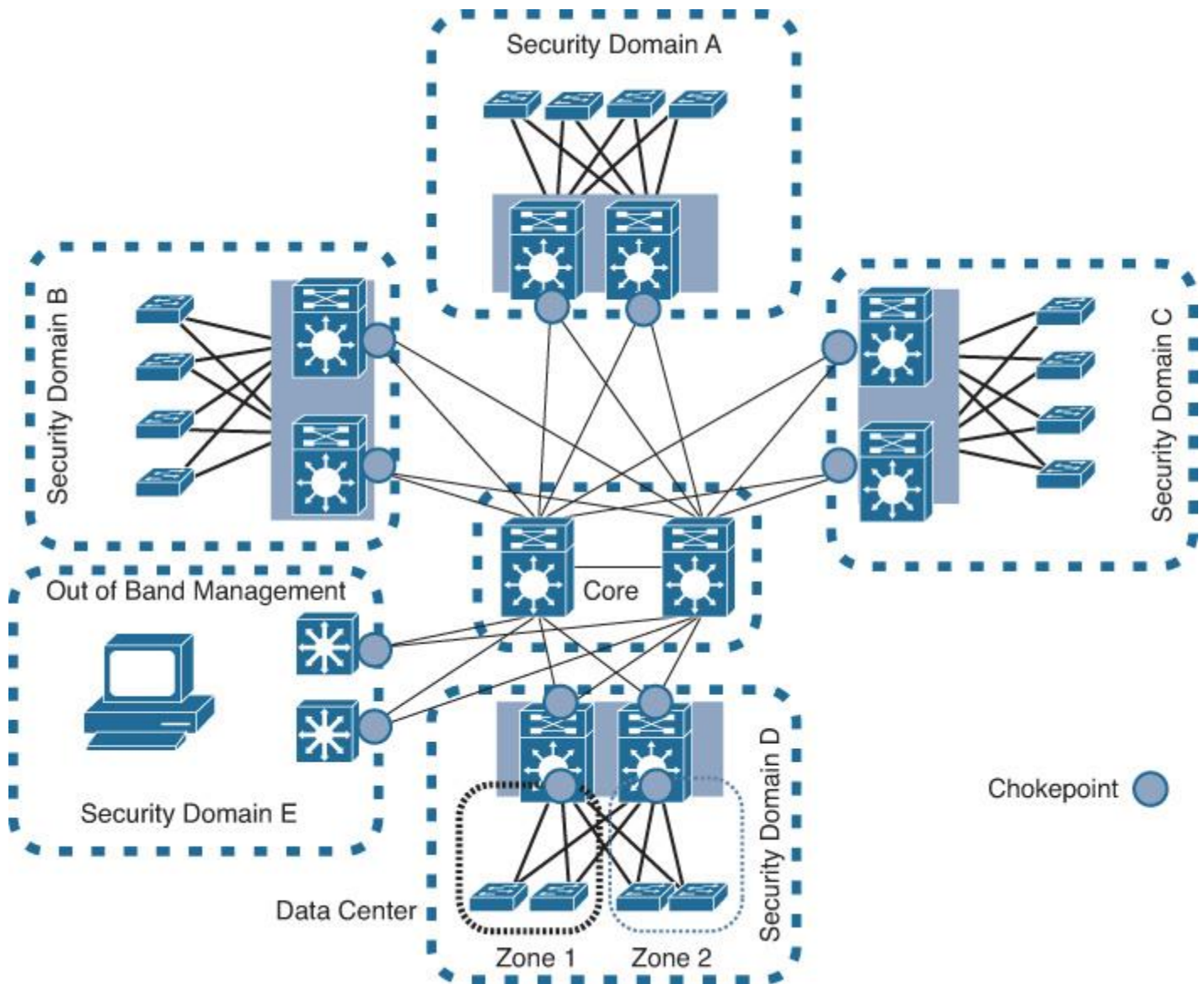


Figure 10-47 Security Domains and Zones

Note

In [Figure 10-47](#), the core block was not placed in its own domain to illustrate one of the recommended and proven design considerations that offloads any additional processing from the network core, such as security policies. However, some design requirements require the core to be treated as separate security domain.

Security devices are almost always placed at the chokepoints (domains or zone boundary points). However, the type of security devices and roles can vary based on the domain or zone to be protected and its location in the network. For example, in [Figure 10.47](#), a firewall is placed at the network management zone boundary to fulfill packet filtering and inspection requirements for OAM traffic flows in both directions, while at the public demilitarized zone (DMZ), there are multiple specialized security nodes such as web application firewall data loss prevention and IPS.

Nevertheless, it is critical that network architects and designers consider the type of the targeted network and its high-level architecture in terms of whether it is an SP network or enterprise network. Each of these networks (irrespective to its detailed design) has different traffic flow characteristics. For instance, enterprise networks always define chokepoint boundaries such as the Internet edge or connections to extranet networks where traffic is always controlled by strict and confined rules. In contrast, as covered earlier in [Chapter 5](#), SP networks are transit types of network where traffic should be allowed to traverse the SP network between different customers' remote sites or peering ISPs in different directions.

By taking these high-level traffic pattern differences and applying the domains or zoning concept to each of these networks, it is obvious that SP networks have many external connections. This translates into many

domain boundary points, usually with either very limited or no specialized security devices positioned at these boundary points with their customers. For example, with MPLS VPN providers who offer L3VPN and L2VPN, the top priority for them is usually to forward the high volumes of their customers' traffic with the least possible latency.

Enterprise networks, however, normally have limited domain boundary points to other domains (outsides domains, such as Internet and links to business partners). Usually these domain boundary points are controlled by several network security devices, with each focusing on a different security layer and function. In other words, although the concept of domains and zoning is applicable to every network, the type of the targeted network influences how this concept can be applied to a large extent.

Security Triad Principle (Confidentiality, Integrity, and Availability)

Confidentiality, integrity, and availability, also known as *CIA*, are considered the fundamentals of information security. In other words, these elements help to construct secure systems that protect information and resources (devices, communication, and data) from unauthorized access, modification, inspection, or destruction. Therefore, any breach to any of these three elements can lead to serious consequences. Consequently, network designers must be aware of these elements and how each impacts the network design, and at the same time how the network design can accommodate these principles as an integral part of it. Table 10-14 summarizes the characteristics of each of these elements and the common mechanisms used to achieve them.

CIA Triad

Security Element	Characteristic	Mechanisms to Achieve
Confidentiality	Protect against unauthorized access to information to maintain the desired level of secrecy of the transmitted information across the internal network or public Internet (in other words, identifying who should have access to this information).	Cryptography, encryption, user ID and password (two-factor authentication is more common today) or security tokens. Cryptography can provide data confidentiality by modify the data into a format that can be understood only by the authorized entity.
Integrity	Maintain accurate information end to end by ensuring that no alteration is performed by any unauthorized entity.	Cryptography, data checksum. Cryptography enables the receiving entity to verify whether there was any alteration to the original, such as hashing.
Availability	Ensure that access to services and systems is always available and information is accessible by authorized users when required. Also, in modern networks, availability is not only measured by network or service up or down, but also by its quality. (For example, malicious traffic may increase latency in the network, which leads to a degraded VoIP quality [users might not be able to make voice call even though the voice system and the network are up]).	Systems designed with high availability at different levels (network, storage, compute, application) and that protect against attacks such as a DoS that attempts to stop access to resources.

Table 10-14 Confidentiality, Integrity, and Availability

Network Infrastructure Security Considerations

Securing today's modern networks is one of the most complicated tasks. Mobility and communication over the Internet are becoming one of the primary (de facto) methods of communications and essential requirements for many businesses. Therefore, to address these trends and requirements, sophisticated security countermeasures are required. Typically, a good security design follows a structured approach that divides the network into domains and applies security in layers, with each layer focusing on certain types of security requirements and challenges. This is also known as *defense in depth*, where multiple layers of protection are strategically located across the network and where a failure of one layer to detect an attack or malicious traffic flow will not leave the network at risk. Instead, the multiple security layers work in a back-to-back manner to detect the attack or malicious flow in the network [99]. [Figure 10-48](#) summarizes the common security aspects that can be applied in a layered approach.

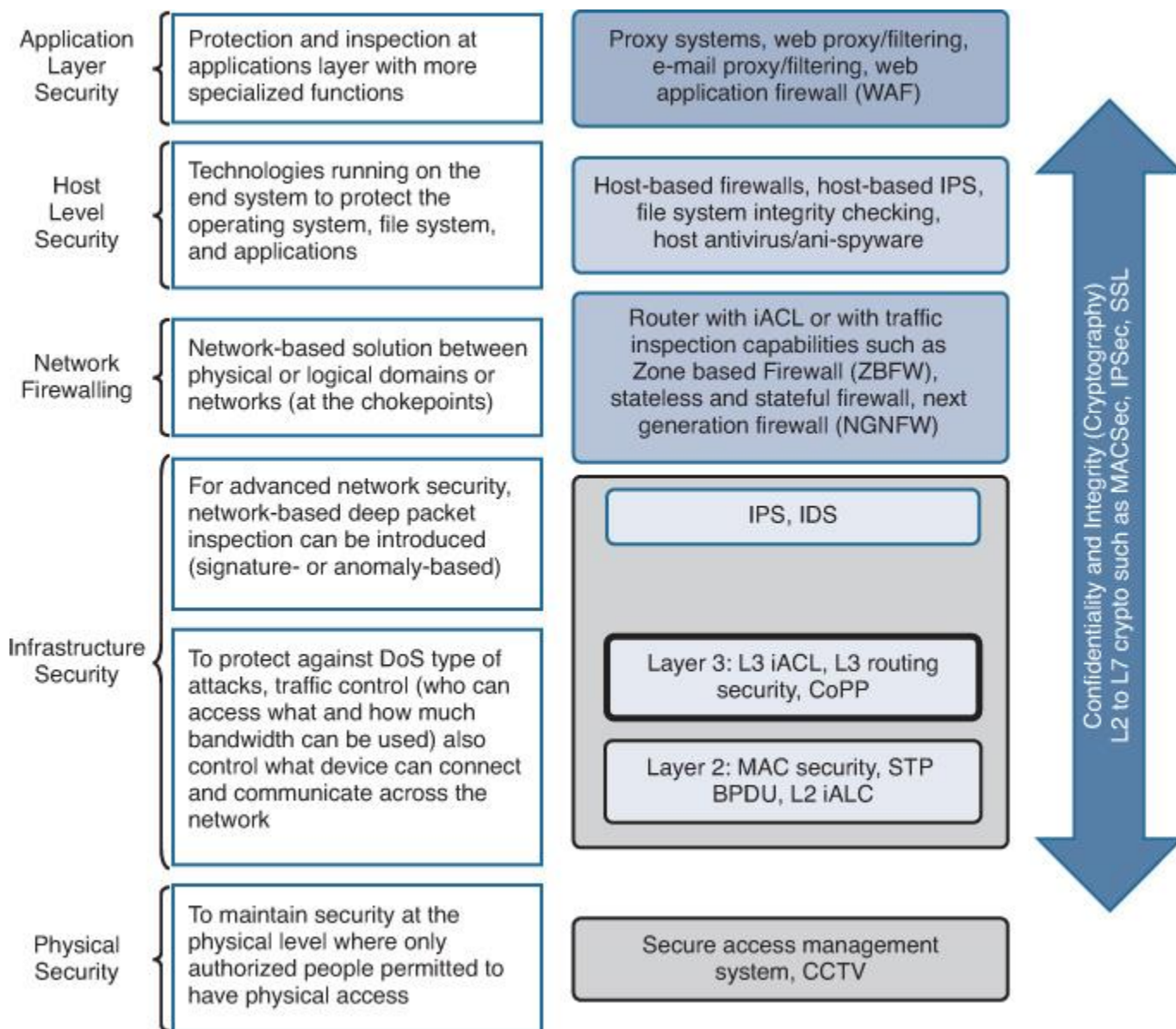


Figure 10-48 Elements of Layered Network and Information Security

This section covers infrastructure security and network firewall considerations, in brief, focusing on the integration and impact with regard to network design.

Network Device Level Security

Securing network devices is the first and essential consideration to protect any network. For instance, even though you may have a very secure control plane and network edge design and policies, if one or two network devices are compromised, the network can be taken down easily by introducing black-holing routes, generating a large amount of malicious traffic that is technically sourced from an internal (trusted) network device, or the attacker can even take advantage and access other internal zones that are not accessible externally. In addition to that, if someone can access a network device, it means all the traffic passing through this device can be sniffed and copied. This is dangerous for an organization that has sensitive information, such as credit card numbers or people's personal information. Therefore, protecting network devices is a fundamental requirement to achieving a secure network design.

To approach device-level protection in more structured manner, as a network designer you need to understand the following types of traffic that network devices normally handle, as summarized in [Figure 10-49](#). Understanding the different traffic types passing through any network will help to identify how each traffic type can impact the device and the possible countermeasures for each.

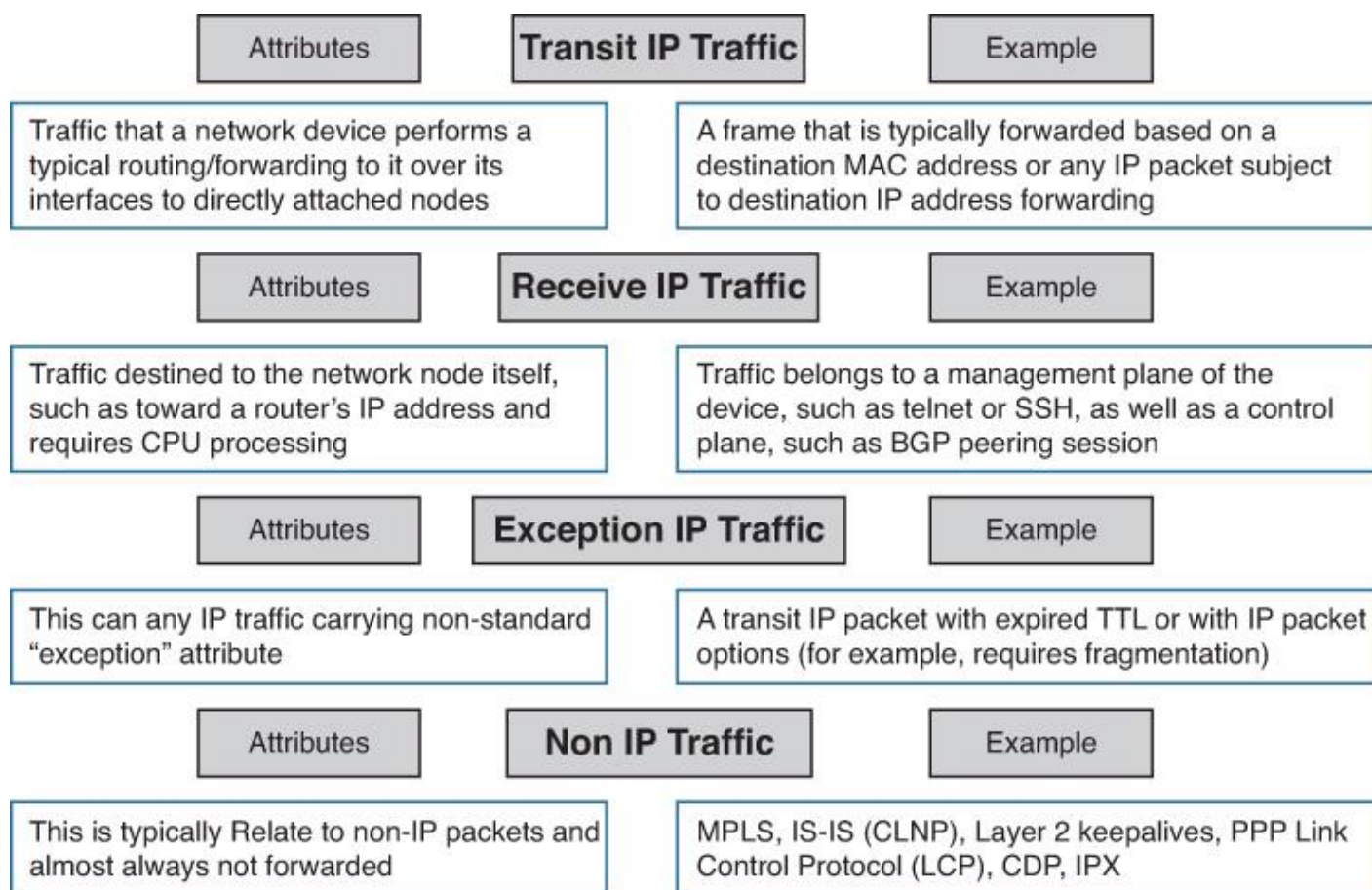
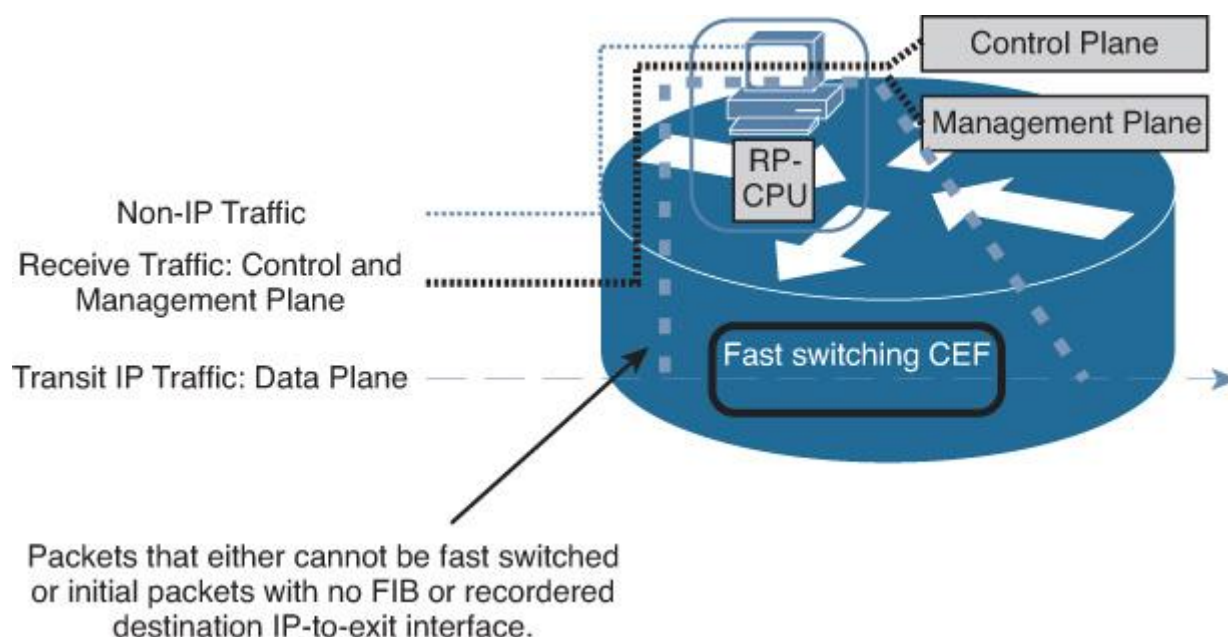


Figure 10-49 Types of Traffic Handled by Network Devices

The different traffic types listed in [Figure 10-49](#) each has impact in terms of overloading network devices with regard to the IP traffic planes covered in [Table 10-15](#) and [Figure 10-50](#), because each has its own attributes and security considerations, based on Cisco's Network Foundation Protection (NFP) framework [100, 120].

Device Plane	Security Concern Example
Data plane	Responsible for controlling fast forwarding of traffic passing through a network device. However, this plane is normally limited to a certain number of packs per second (pps) based on the hardware platform's throughput. iACL, QoS toolset, remotely triggered black hole (RTBH), and Unicast Reverse Path Forwarding (uRPF) are common mechanisms to protect this plane.
Control plane	As discussed earlier in this book, the control plane is like the brain of the network node; it usually controls and handles all Layer 3 functions. Therefore, any control plane-related issues such as a flapping session with a BGP peer that advertises an extremely large number of prefixes will not only impact the network stability, but also the device itself will be impact because of high CPU spikes in this case. iACL, routing protection, and control plane policing (CoPP) are common mechanisms to protect this plane.
Management plane	As the name implies, this plane relates to the management traffic of the device, such as device access, configuration, troubleshooting, and monitoring. Therefore, its criticality is equal to the other planes described earlier. Any unauthorized access can lead to a device and network-wide crisis, such as injecting a black-holing route into the control plane or flooding the network with malicious traffic, which will ultimately impact all the hosts and users transiting though the network in general and this network device in particular. CPU and memory thresholding, AAA (authentication, authorization, and accounting), and CoPP are common mechanisms to protect this plane.

Table 10-15 IP Traffic and Device Planes



Taking the different traffic types and how each is mapped into the relevant plane for processing in a structured manner, the following points should be considered as general and foundational guidance to achieving the desired level of device hardening and security level.¹³ Consider that these points should not conflict or breach the security policy or company standards.

13. “Cisco Guide to Harden Cisco IOS Devices,” <http://www.cisco.com>

■ **Physical security:** The first basic security consideration is that network devices be placed physically in a secure place, where only authorized people can access them.

■ **Authentication, authorization, and accounting:** Network devices must be accessible only by authorized people. Ideally, the access privileges must be multilevel role-based access (authorization level “who can do what”) combined with a reliable accounting (logging) to keep track of who has access the devices and what has been done.

■ **Secure the device management plane:** Ideally, the management sessions and activities of the network devices must be performed over protected protocols such as Secure Shell (SSH), Secure Sockets Layer (SSL), Secure File Transfer Protocol (SFTP), or IPsec. Also, it is always advised that an out-of-band management access to network devices be provisioned to offer secure, guaranteed direct access to critical network devices in situations where the network might be facing a DDoS attack (because in-band management access will be almost impossible).

■ **Protect the hardware resources:** Maintaining a device hardware recourse is the foundation to maintaining traffic flow through the device. Therefore, protecting hardware resources such as CPU is an essential countermeasure to be considered with regard to device hardening, such as CoPP to protect the CPU from spikes because of excessive control plane traffic. Control plane protection (CPPr) also helps to protect from other types of traffic, such as exception and transit traffic.

■ **Disable unused services:** A network node does not usually use every single service it has; normally, the device role and function within the network dictate which required services need to be turned on. In general, unused services should be disabled to limit the chances that an intruder or hacker can compromise one of these unused services to generate malicious traffic or DoS attacks.

■ **Monitoring:** Network security is dynamic and always evolving and changing in response to the continuous technology and design developments and changes. As a result, there are always new system vulnerabilities, breaches, and intrusion developments. Therefore, to keep track of what is going on in terms of device utilization (bandwidth or hardware resource) and whether it is within the normal (baseline) limits, network and device monitoring helps network operators to always optimize device security either proactively or in a fast, reactive approach. Simple Network Management Protocol (SNMP) and NetFlow are the most common protocols used for this purpose, but not the only ones.

Although the device-level security considerations covered earlier are considered generic and standard foundational guidelines, the consideration of these guidelines will vary based on some other factors. Figure 10-51 identifies the primary influencing factors with regard to device-level security.

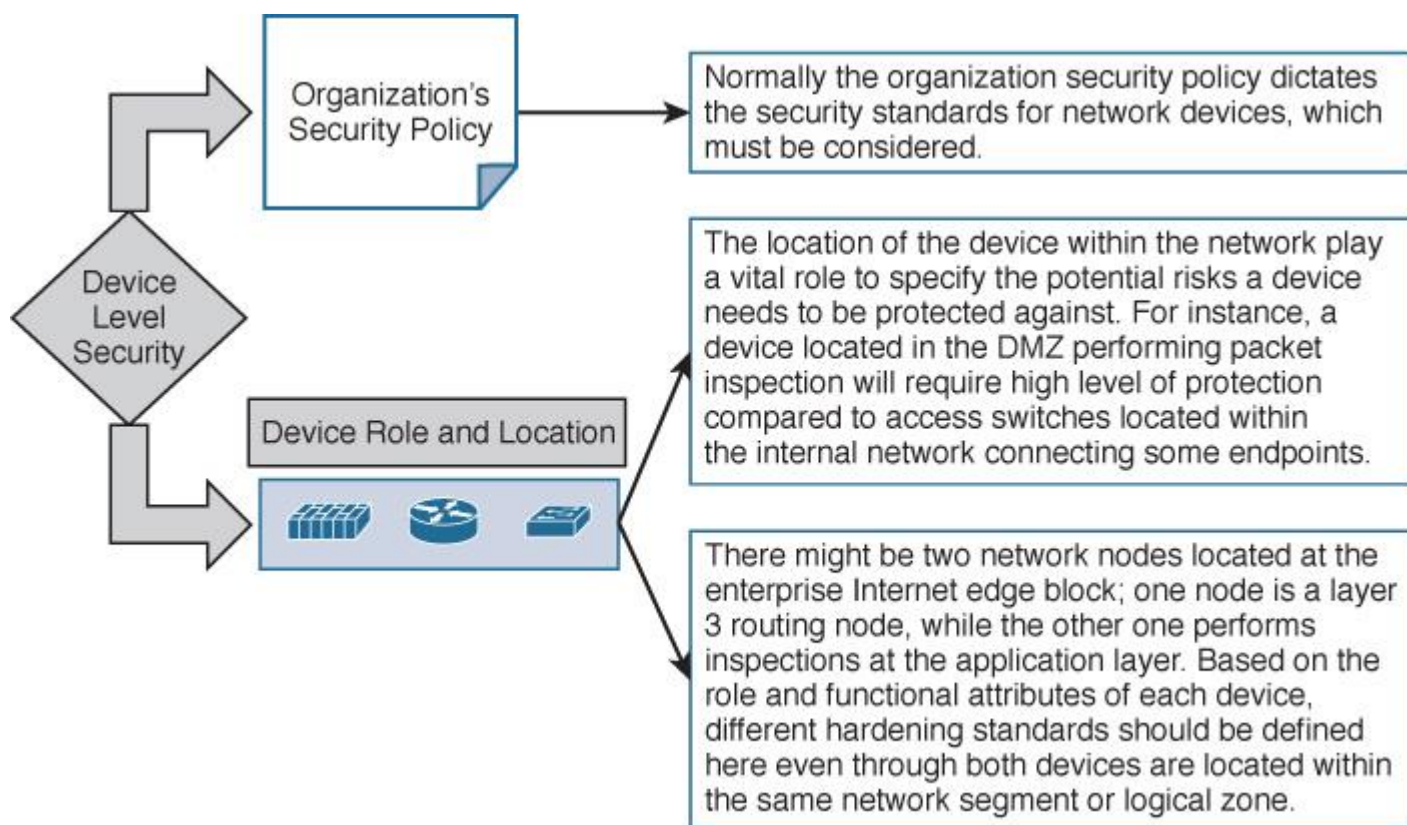


Figure 10-51 Influencing Factors on Device-Level Security

Layer 2 Security Considerations

Securing Layer 2 or data link layer communication is as critical as other higher layers in the network. One of the common practices is that network designers or operators focus more on securing higher layers across the network, such as the Layer 3 control plane, protocol, and session layers, without giving Layer 2 enough focus to secure it. As covered in multiple sections in this book, a building constructed with a weak foundation will not be able to deliver the promised value or end result and will be more vulnerable to damage.

Similarly, a highly secure network on the higher layer, such as network and application layers, can be easily taken out of service by simple Layer 2 flooding across its switched network. Therefore, it is critical to secure the network in a bottom-up manner to achieve a reliable and self-healing network security design. Layer 2 compared to other OSI layers is the simplest with regard to security considerations and one of the most dangerous at the same time. For instance, if a hacker compromises a data center switch, he can easily sniff all the traffic passing through this switch that might carry critical data. At the same time, this hacker can simply take the data center switched network down by flooding the ternary content-addressable memory (TCAM) table of the switch or sending malicious Layer 2 traffic storms. In addition, IPv6 raises a number of first-hop security (FHS) concerns that were not present in IPv4. Layer 2 security can be categorized as the following to produce a structured and flexible Layer 2 security design:

■ **Authentication and authorization:** Normally, the Layer 2 switched network is the network edge or boundary where end users and hosts connect. Typically, it is the edge where rogue and unauthorized endpoints can be connected. Therefore, only authorized devices must be permitted to connect to the network at the switch port level. This normally can be achieved by using different mechanisms and features, such as authentication with IEEE 802.1X, which can integrate with a network admission control system to authenticate users and their endpoints, port ACL (routed ACL, MAC ACL, VLAN ACL), DHCP snooping, and other port security features. (One or a combination of these mechanisms or features can be used based on the security goals, standards, and supported features. Also, IPv6 router advertisement (RA) guard, DHCPv6 guard, and IPv6 snooping help to mitigate IPv6 FHS concerns. Furthermore, the IEEE 802.1AE standard known as MACsec can be used to satisfy some security specifications that require protecting the data traversing an Ethernet LAN such as between the Layer 2 DCI edge nodes of a secure data center in which MACsec offers

the ability of authenticating and encrypting packets between the two MACsec-capable devices at the DCI edge of the interconnected data centers.

■ **VLAN design:** Although one of the primary goals of VLANs is to provide separation at Layer 2, the design of these VLANs can sometimes lead to security concerns. For example, if an extended VLAN across two data centers over a DCI is compromised, this may lead to a major impact and risk to both data centers. Therefore, it is always recommended that Layer 2 domains should be contained to a limited range within the network (smaller failure domain); this also avoids issues such as Address Resolution Protocol (ARP) or unicast flooding. However, practically this is not always achievable because the reality is that application requirements dictate how the network should be designed. Accordingly, protective features and policies have to be considered in scenarios like this, such as defining rate limiting along the path and mitigating the impact of the traffic flooding. In addition to the typical separation VLANs provide to the design, private VLANs (PVLANS) offer a more advanced level of a controlled traffic separation between VLANs and between endpoints that reside within the same VLAN. Furthermore, the use of VLAN Trunking Protocol (VTP) can introduce a high risk to the switched network because inserting a switch that takes over the VTP server role will rewrite VLANs set up across the entire switched network under the same VTP domain. Therefore, securing VTP or considering VTP in transparent mode is always recommended to avoid its impact in situations like this.

■ **Layer 2 control plane:** As discussed earlier in this book, Layer 2 control protocols, such as Spanning Tree Protocol (STP), provide protective mechanisms against Layer 2 loops. However, STP is considered a double-edged sword because in the case of misconfiguration, with any security weakness any attacker can simply bring down the switched network or change its logical design. This is because no reliable authentication capability is built in to the STP. Therefore, if someone attaches a switch to the existing switched network that has better attributes than the other switch to make it the network root bridge, the network first will be unstable for a while during the election process of the new root bridge. Second, after this rogue switch is considered as the root, a disaster will probably occur in this network because of performance issues, and may introduce traffic black-holing at Layer 2. To further exacerbate the situation, most of the traffic will pass through this new root switch in which anyone can sniff the traffic or take it or copy it. Moreover, this behavior can be dangerous to metro Ethernet providers if one of their customer switches become the root bridge. Therefore, protecting the Layer 2 control protocols used is an essential consideration to secure Layer 2 switched networks, such as protecting the root bridge by denying any other switch in the network from participating or sending a rogue message to be elected as a root bridge and filtering Layer 2 bridge protocol data unit (BPDU) messages at the network edge as well.

Layer 3 Control Plane Security Considerations

The Layer 3 control plane is the intelligence over the network that steers traffic toward reaching its intended destination. This means any wrong information can lead to traffic black-holing and the eventual dropping of packets. In addition, in a network with unprotected routing, anyone can easily bring it down by either flooding a large number of prefixes (legitimate or not) by inserting a rogue router or redirecting traffic a black-holing next hop or a sniffer. Therefore, securing the Layer 3 control plane is a must. However, the question here is this: Where is securing a control plane a must, and where is it just something beneficial to have? Although there is no one standard answer to this generic question, the following can be considered as general rules with regard to Layer 3 control plane security:

■ Secure the control plane against rogue Layer 3 peers by authenticating routing sessions. This is a recommended practice to be considered across the entire routed domain and strongly recommended when peering with an external routing domain.

■ Device protection with regard to Layer 3 control plane functions is a primary element to be considered. For example, if a router that is not supposed to receive and process prefixes receives a large number of them, this node might run out of memory. In all likelihood, its CPU will be overloaded, and this will result in a malfunctioning node in the routing path, which will introduce instability (because the routing session will probably keep going up and down in this case). A large percentage of traffic will also be dropped. Therefore,

it is important to limit the maximum number of routes to be accepted by the node. This is common in MPLS L3VPN SP design, where normally the PE nodes limit the maximum number of prefixes per VRF/VPN.

■ Accept only the routes that are expected to be received. This is common in MPLS L3VPN and ISPs, where the ISP, for example, will only permit the Provider Assigned “PA” or Provider Independent “PI” address space the customer has to advertise with the agreed subnet length, sometimes combined with uRPF check. For instance, a customer who owns /24 IPv4 public ranges will not normally be allowed to divide it and advertise it in the format of host routes (/32) unless it is agreed with the ISP. This is because the ISP always tries to protect their PE nodes from being overloaded with extra thousands or even can reach hundreds of thousands of extra routes.

■ Advanced BGP techniques such as QoS policy propagation with BGP (QPPB) and remote black-holing trigger can be used in SP-style networks to offer a more dynamic, flexible, controlled, and secure control plane design that is almost impossible to be achieved with the typical static filtering mechanisms, such as using ACLs or prefix lists.

■ QoS has powerful features that help to optimize Layer 3 control plane security design, usually by guaranteeing a minimum amount of bandwidth during congestion situations. This ensures the routing session will not be torn down, and it limits the amount of bandwidth to a predefined maximum allowed bandwidth. This protects the network from undesirable behavior such as DoS attacks by a rogue router, for example, that generates a large amount of traffic using the same control plane marking or even TCP/UDP port. To take advantage of this QoS as a protective mechanism here, it ideally should be applied across the entire routing domain and not only at the edge of the network.

■ CoPP, as covered earlier, should be considered on every network node to relax the impact of control plane flooding and protect network nodes from CPU spikes by blocking unnecessary or DoS traffic.

Note

Although CoPP uses modular QoS CLI (MQoC), the QoS considerations earlier provide network-wide treatment on a per-hop basis for Layer 3 control plane traffic flows, whereas CoPP is focused only on a device level. As covered earlier, marking down the DSCP value of packets that are out of profile can impact how these packets will be treated by other nodes across the network.

Remote-Access and Network Overlays (VPN) Security Considerations

The most common and proven mechanisms for providing secure remote access for remote users and remote sites over the public Internet is the overlaid virtual private network (VPN). One common misconception is that the term *VPN* is always seen as a synonym for IPsec. In fact, IPsec offers cryptography at the network layer, whereas VPN can take different forms, such as MPLS L3VPN, MPLS L2VPN, SSL VPN, DMVPN, and so on. Therefore, it is important that network architects and designers distinguish between VPN and IPsec. Having said that, secured VPN with IPsec is the most common and proven combination that is used to protect the various types of VPN solutions. [Table 10-16](#) compares the most common overlay VPN solutions from different design aspects.

	IPsec	GRE	DMVPN	GETVPN	Remote Access (Client Based)
Standard	Yes	Yes	No (Cisco proprietary)	No (Cisco proprietary)	Yes
Overlay transport	P2P IPsec tunnel	P2P GRE	mGRE	Tunnel-less	P2P IPsec tunnel
Protection model	Peer to peer	Peer to peer with IPsec	Peer-to-peer with IPsec	Group protection (RFC 3547)	Peer to peer
IP routing	Static routing	Tunneled dynamic routing	Tunneled dynamic routing	Standard IP WAN dynamic routing	Reverse-route injection
Offers a protected VPN over public Internet	Yes	Yes, with IPsec	Yes, with IPsec	No	Yes
Offers a protected VPN over the IP WAN transport	Yes	Yes, with IPsec	Yes, with IPsec	Yes	Yes
Requires some changes to the existing Layer 3 routing design when used over the WAN	Yes	Yes	Yes	No	Yes*
Requires additional network equipment	No	No	No	Yes, such as key server	No
Support NATing	Yes	Yes	Yes	No (NAT must be performed before encryption or after decryption.)	Yes

*This VPN is commonly used to provide connectivity over the Internet.

Table 10-16 Overlay VPN Solutions Comparison

One main consideration with regard to an enterprise VPN is whether to enable the VPN across the enterprise WAN. The typical answer to this depends on the business requirements, security policy, and the functional requirements. One must consider the impact on the business-critical applications because of the increased packet overhead by the VPN tunneling, especially when VPN is combined with IPsec, as shown previously in [Figure 10-38](#). Besides, it is common that security devices in the path such as firewalls with NATing add a layer of design and operational complexity to the VPN solution.

In addition, introducing a VPN across the WAN can increase the design and operational complexity in scenarios where dual WAN connectivity is required (either dual routers or links). In these cases, the routing redundancy design will be moved from the typical IGP or BGP to IPsec redundancy, which might not be a suitable option for some design requirements. Therefore, GETVPN is becoming the most common and desirable secure VPN solution of the WAN because GETVPN by nature offers tunnel header preservation, which facilitates routing encrypted packets using the underlying network routing infrastructure. There is no need to consider any redundancy design at the VPN level, such as dual hubs or stateful IPsec high availability.

Even though securing IP traffic across the WAN will introduce a new layer of complexity, secure IP communication is a top priority for some businesses, like those in financial services environments. Sending internal traffic across the WAN will be seen by the business as trusting SP setup, security, employees, and even contractors and third parties who work and integrate with this SP not to perform any malicious activity. Therefore, for this type of business, it is common that IPsec, such as secure DMVPN or GETVPN, is deployed across the WAN.

In other words, enabling secure IP communication over the WAN is not a common practice because it is seen as adding an unnecessary layer of complexity and may impact the performance of some business applications, whereas for those business that are concerned most about security, secure IP communication over the WAN is essential. Therefore, the decision of whether to enable secure IP communications across the WAN is almost always based on the business type, priorities, and impact on the overall network and application performance, in addition to operational complexity.

Network-Based Firewall Considerations

Technically, a firewall is not a device built to perform routing or switching. Instead, it is a device that is primarily intended to perform security functions such as packet filtering, inspection, and VPN termination. Although almost all the firewalls nowadays support static and dynamic routing, the nature of these devices almost always impacts the network design. This impact varies based different factors and attributes. The following are the primary and most common and influencing factors that network designers must always consider whenever there is a firewall in the network path:

■ **Stateful versus stateless firewall:** Typically, stateless firewalls deal with traffic in a static manner where traffic flows are only monitored and filtered (allowed or blocked) based on static information such as IP source, IP destination, or port numbers. This behavior from a network design point of view can introduce limitations and operational complexities because every single traffic flow needs to be included in both directions (ingress and egress) as part of the firewall entry rules. Stateless firewalls are also not very friendly with applications that automatically negotiate ports, such as RTP (UDP) ports, which are normally automatically negotiated during the signaling and establishment of a VoIP or video call. Stateful firewalls, in contrast, offer traffic monitoring and filtering that is more aware of traffic flows to overcome the limitations of the stateless firewalls, especially with applications that require dynamic port negotiation to establish their session, such as FTP, VoIP, and video RTP media streams.

■ **Stateful versus stateless firewall failover:** Even though some Firewalls work in active-standby mode are stateful firewalls from a traffic handling perspective, the way these firewalls fail over from the active to the standby can take two forms, either stateless failover or stateful failover. With the stateless failover, the state table is usually not replicated to the standby peer firewall. This means in the event of active firewall failure, the active role will be moved to the standby firewall. The issue here is that after the failover happens, all the connections (active flows) have to be reinitiated. In contrast, stateful failover offers the ability to replicate the state table between the active and the standby firewalls to avoid reinitiation of the active session

following a failover event. The impact of a session's reinitiation might not be an issue for some applications and businesses, but it can be a serious issue for others. Therefore, the decision whether to use stateless or stateful failover should ideally be driven by the design and application requirements.

■ **Routed versus transparent firewall modes:** The other important factor that has a direct impact on network design with regard to firewalls, apart from the operational attributes of the firewalls (stateful or stateless), is that firewalls can operate as a Layer 3 or Layer 2 node. Technically, from a network design point of view, introducing a Layer 3 node into a routed network means that there will be a need to change IP addressing or a need for new IPs in the area where this firewall will be added, in addition to considerations about whether the firewall supports dynamic routing. For instance, if a firewall is added to an Intermediate System-to-Intermediate System (IS-IS) routed domain and the firewall does not support IS-IS, in this case static routing will be considered. This means a redesign of the routing will be required. Firewalls operating in Layer 2, however, will not introduce these challenges and can be inserted transparently into the routing domain. This is commonly known as *transparent mode* because there is no change required to the existing routing, IP addressing, or host default gateways. Furthermore, when a firewall operates in a transparent mode, it often loses some of its functions or features, such as providing VPN termination. However, this is completely vendor and device model dependent and can vary from one to another to a large extent. Remember that the CCDE exam is not hardware or vendor dependent. Therefore, if something relates to nonstandard software or hardware specification, it should be provided as part of the design scenario.

Based on the multiple sections and security layers discussed briefly in this section, [Figure 10-52](#) provides a summarized list of some common network attacks and risks, along with the possible countermeasures and features that can be used to protect the network infrastructure and mitigate the impact of these attacks at different layers.

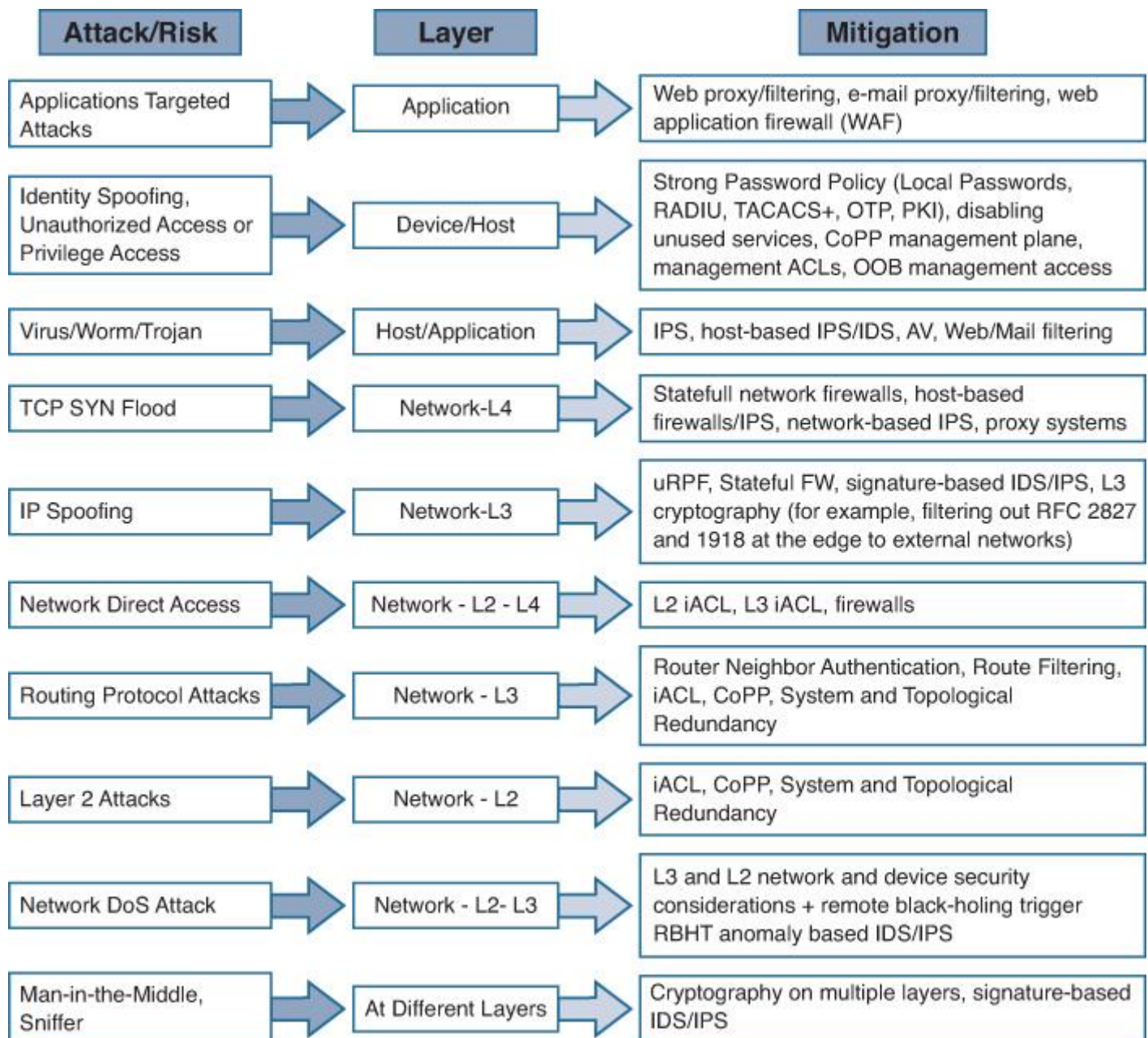


Figure 10-52 Common Network Attacks and Mitigation Mechanisms

Further Reading

RFC 6480, *An Infrastructure to Support Secure Internet Routing*, <http://www.ietf.org>

RFC 2725, *Routing Policy System Security*, <http://www.ietf.org>

RFC 7416, *A Security Threat Analysis for the Routing Protocol for Low-Power and Lossy Networks (RPLs)*, <http://www.ietf.org>

RFC 4272, *BGP Security Vulnerabilities Analysis*, <http://www.ietf.org>

RFC 7454, *BGP Operations and Security*, <http://www.ietf.org>

RFC 3948, *UDP Encapsulation of IPsec ESP Packets*, <http://www.ietf.org>

"Cisco Guide to Harden Cisco IOS Devices," <http://www.cisco.com>

"Cisco SAFE Reference Guide," <http://www.cisco.com>

Network Management

As discussed earlier in this book, today's modern networks carry multiple business-critical applications over one unified infrastructure such as voice, video, and data. In addition, in today's competitive

telecommunications market, SPs always aim to satisfy their customers by meeting strict SLA requirements. As discussed earlier, various technologies, protocols, architectures, and constraints all collectively construct an operational network. However, designing and deploying a network using a business-driven approach will not guarantee the quality of the solution as well (that is, if the network is really operating as expected or not).

Moreover, traffic requirements in terms of pattern and volume can change over time as a natural result of business organic growth, merger and acquisition, and the introduction of new applications. This means that the network may end up handling traffic flows that it was not designed for. The question here is this: How can IT leaders and network operators know about what is going on? If the network is facing performance issues that need to be taken care of, how can the change and alteration be performed in a tracked and structured manner? In fact, configurations and changes are more of a concern compared to other aspects because any error can lead to either a downtime or a degraded performance, and based on recent studies such as the one conducted by ZK Research in 2013 with regard to network management, 37% of network downtimes are caused by human error, which makes its impact the biggest compared to other influencing factors, as shown in the [Figure 10-53](#) [118,119].

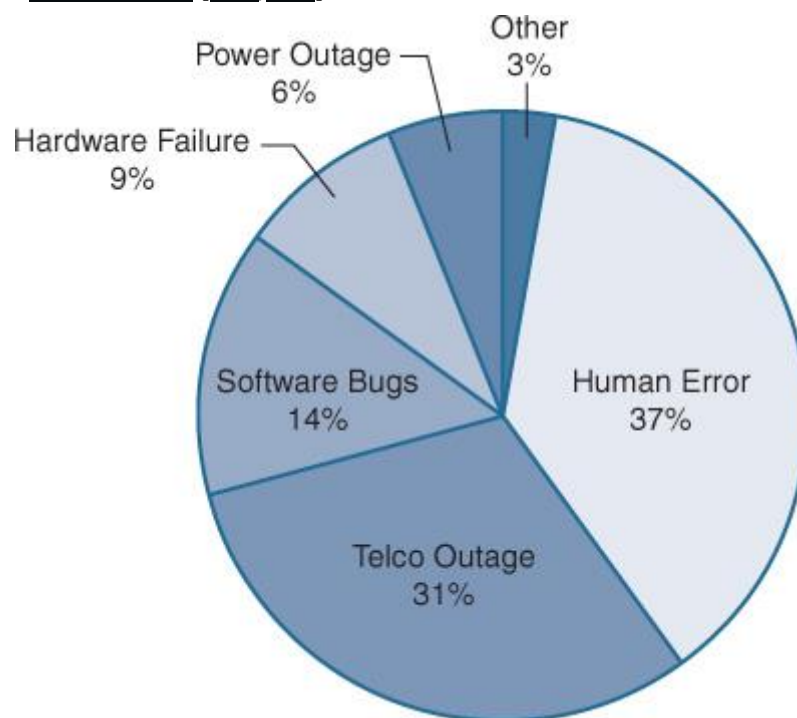


Figure 10-53 Causes of Network Downtime Based on ZK Research, 2013

Consequently, there must be a set of procedures and mechanisms capable of measuring and providing real-time and historical information about every single activity across the network to help the IT team take action in a more proactive manner instead of relying on only the reactive approach to be able to effectively, and in a timely manner, identify and fix issues or abnormal behaviors in which the mean time to repair (MTTR) needs to be kept as short as technically possible. Furthermore, the action taken by IT should also be performed in a controlled and structured manner to be tracked and recorded, also combined with some automation with regard to configuration and changes to help reduce the percentage of human errors. For the IT to achieve this, they need a network management solution that controls operation, administration, maintenance, and provisioning [88].

There are several industry standards and frameworks in the area of network management. This section discusses the ITU-T standard (FCAPS).

Fault, Configuration, Accounting, Performance, and Security

FCAPS is a network management framework defined by the International Organization for Standardization (ISO) to help organizations classify the objectives of network management into the following five distinct layers or categories:

■ **Fault management:** This management layer aims to minimize network outages by employing a set of procedures and activities to detect and isolate network issues, along with the appropriate corrective actions to overcome current issues and prevent them from occurring again. Alarms, fault isolation, testing, and troubleshooting are example of fault management functions.

■ **Configuration management:** This management layer aims to maintain a current inventory of network equipment, with its configurations to be used for planning, installation, and provisioning of new services and network equipment.

■ **Accounting management:** It is implied by the name that this layer aims to ensure that each user or entity is billed or allocated an appropriate cost reference on the activities and utilization performed across the network. This can be measured based on various elements, such as a given service usage or bandwidth utilization. Usage management, pricing, auditing, and profitability analysis are examples of accounting management functions.

■ **Performance management:** This layer aims to monitor and keep track of any performance issues, such as network bottlenecks, by continuously collecting and analyzing statistical information to monitor, correct, and optimize any reduced responsiveness across the network. This will potentially lead to enhanced capacity planning and quality measurement. Quality assurance, performance analysis, and monitoring are examples of the functions of this management layer.

■ **Security management:** Typically, this layer focuses on the security of the management solutions (all the different layers above) in terms of access control, data confidentiality, and integrity with regard to network alarms, events, and reports. In addition, this layer sometimes refers to the monitoring and management of the network with regard to the security aspects such as unauthorized access, traffic spikes (DoS attacks), and targeted applications attacks.

Network Management High-Level Design Considerations

Network management is an extensive and large topic. In addition to that, network management is defined and approached differently based on the entity or organization that creates the standards and the aim of the standards or framework, such as FCAPS versus ITIL. Therefore, it is impossible to cover such a large and extensive topic in a single section or chapter. This section covers the primary points and considerations at a high level to help network designers to drive the considerations around network management in the most suitable direction. Typically, there can be more than one right direction or approach.

The answers to the following questions help to form the foundation of the network management solution:

- What is the targeted environment (such as enterprise, MPLS VPN SP, application SP, cloud-hosting SP)?
- Is there any existing network management solution? If yes, does the solution follow any standard approach or framework such as FCAPS?
- Is the solution to be added to overcome an existing challenge or for enhancement purposes?
- Are there any business-related constraints, such as budget?
- What is the goal of the solution (for example, monitoring and fault management, capacity planning, billing, monitoring for security purposes, or a combination of these goals)?
- Are there any security constraints with regard to enabling network management, such as only out-of-band management, or can secure in-band management protocols be used?

After all or most of these questions have been answered, network designers should have a good understanding about the high-level targeted network management solution and should be able to start specifying its detailed design, which should answer at least the following questions:

- What information or events do network management solutions need to collect or monitor?
- Where is the best place to gather the intended information or report the relevant events?
- Where should this information or these events be sent after the collection process?
- What is the degree of detail required and is full or partial data collection is required?
- Is the underlying transport network secure (internal) or untrusted (public Internet)?
- How is confidentiality and integrity of the polled or exported information maintained?
- What are the supported protocols and versions by the elements to be monitored and managed for the purpose of network management, such as SNMP or NetFlow?

Taking these questions into consideration, network designers can drive the solution selection and can specify which features and protocols are required and where they should be enabled. For example, an MPLS SP decided to offer VoIP service to their customers to make voice calls to public switched telephone network (PSTN) numbers across the SP IP backbone, in addition to Internet Session Initiation Protocol (SIP) VoIP applications using Cisco Unified Border Element (CUBE) as a SIP gateway, as shown in [Figure 10-54](#).

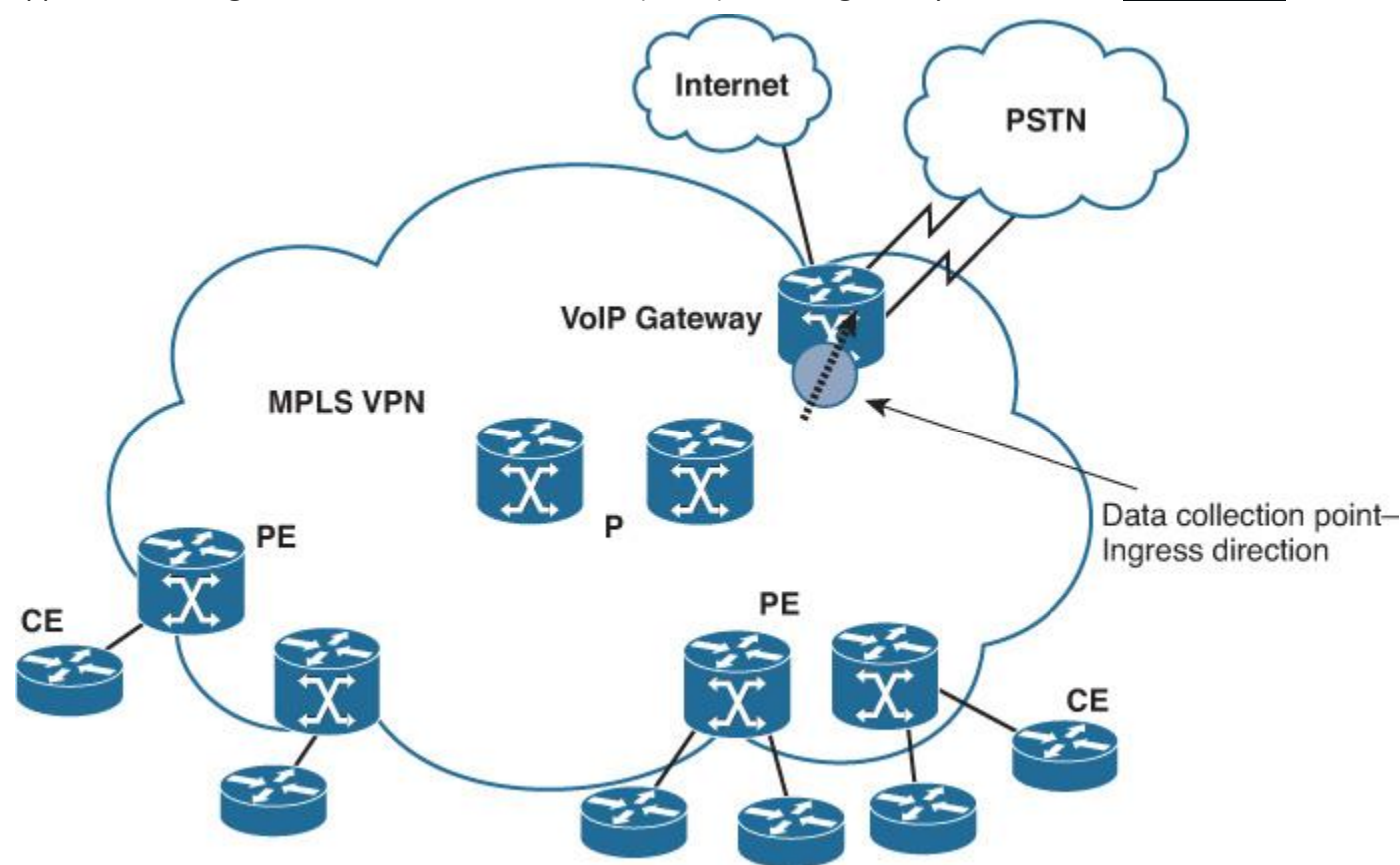


Figure 10-54 MPLS VPN SP Sample Network

According to the nature of this network, solution goals, and architecture, an accounting management needs to be considered here to measure the utilization of the service. In this example, VoIP calls usually require a collection of the actual usage in terms of duration and destination number, which is normally an endpoint IP or another voice gateway IP or CUBE (such as to another SIP URI or PSTN number, local or international), and then a correlation between the collected information and pricing needs to be performed to generate a billing per VPN customer/user per call. By enabling NetFlow at the centralized CUBE, this SP can export VoIP call usage to the NetFlow collector to obtain the desired reports for the billing purposes.

In this particular example, the main question is where to enable data collection. Is it better at the SP edge (PEs), SP core (P), or at the CUBE (voice gateway)? Based on the nature of the service and the traffic flow described, voice calls will be from customer sites toward the services edge gateway, and no VoIP calls are made: neither directly between the different VPN customers nor between sites of the same VPN customer

across the MPLS VPN provider backbone. This means that every VoIP traffic flow will pass through the voice gateway, where the SP can perform ingress accounting per VoIP call flow in a centralized manner.

Another example is an enterprise with hundreds of remote sites connected over a single WAN network that carries different types of traffic, including VoIP, video, and data applications, as shown in [Figure 10-55](#). This enterprise needs a network management solution that monitors WAN link utilization and reports any link that exceeds the available WAN bandwidth with regard to the traffic sourced from each remote site LAN. Based on these simple requirements, it is obvious that a performance management solution is required for the WAN links, where SNMP, for example, can be enabled per WAN link for the network management system (NMS) to collect the relevant Management Information Bases (MIBs) with regard to the utilization of each remote site WAN link. SNMP MIB polling also offers the ability to perform the polling based on a predefined time interval to retrieve the required information only when needed, to reduce the impact on the nodes' CPU and WAN performance.

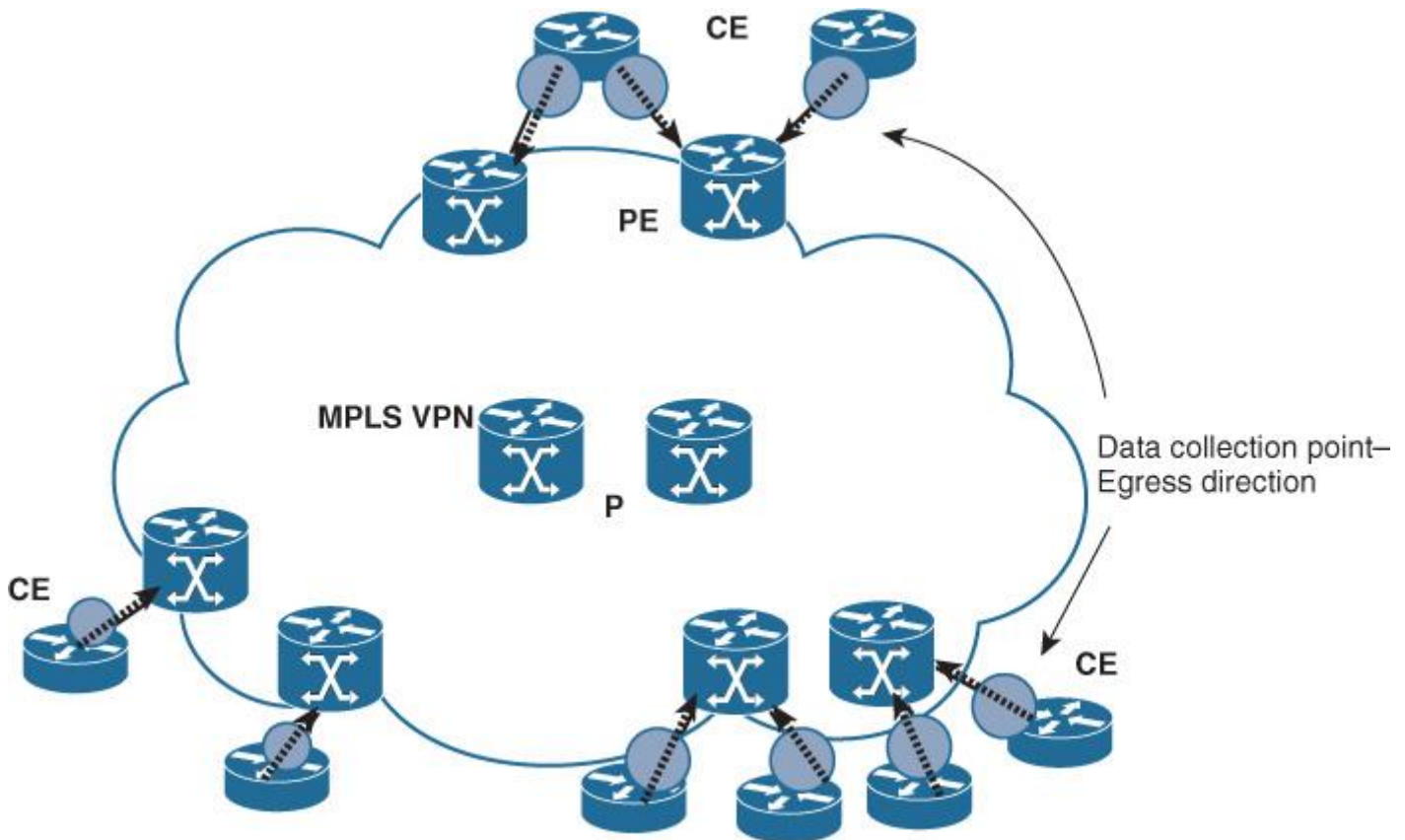


Figure 10-55 Enterprise Network Sample for Network Management

However, if this enterprise requested that the utilization reports have to specify the top five protocols or applications using the WAN link during peak hours and the sources and destinations of these top five traffic flows during peak hours, in this case egress accounting on the WAN links with NetFlow (IPFIX) is required to provide this level of visibility (per application, per session, or flow utilization).

In this example, there are two primary design considerations worth analyzing. The first is the location of the data collection. Based on this scenario's requirements, a performance management solution is required to measure remote site WAN link utilization. This means a distributed data collection model is required here to measure and report the performance of the WAN link per remote site.

The second design consideration is in what direction the data or flows should be metered (ingress versus egress). Technically, in this particular example, to measure a CE router WAN link utilization for traffic sourced from the LAN side, in addition to identifying traffic types and top talkers over this link using different network management protocols (such as SNMP interface counters or NetFlow), measuring traffic in the egress direction will be an optimal choice. In addition, if the link measurement is considered at the egress direction in this example, it is recommended not to consider ingress data collection at the same node for the same purpose, to ensure that there will be no data duplication reported to the NMS.

From these two scenarios, it is obvious that the top-down approach is the most appropriate approach to developing and proposing a network management solution, starting from business goals and working toward the network management protocols.

Multitier Network Management Design

In general, it is proven that integrating and structuring multiple management systems and tools in a hierarchical manner offers a more flexible and efficient network management solution when multiple elements and managements are present. For instance, this layered approach helps to reduce the number of alerts seen by network operations support staff, only presenting filtered and relevant information and alerts. However, to achieve this in a large-scale network, a considerable amount of structured integrations are required between multiple management systems and tools in a layered approach. As shown in [Figure 10-56 \[89\]](#), this approach is recommended by Cisco systems because it offers the following benefits:

- Proactively identifies and corrects potential network issues before they become problems
- Offer optimized IT solution productivity by reducing and eliminating network connectivity loss to a minimum
- Focuses on the solution instead of the problem, which helps to reduce downtime duration (MTTR)

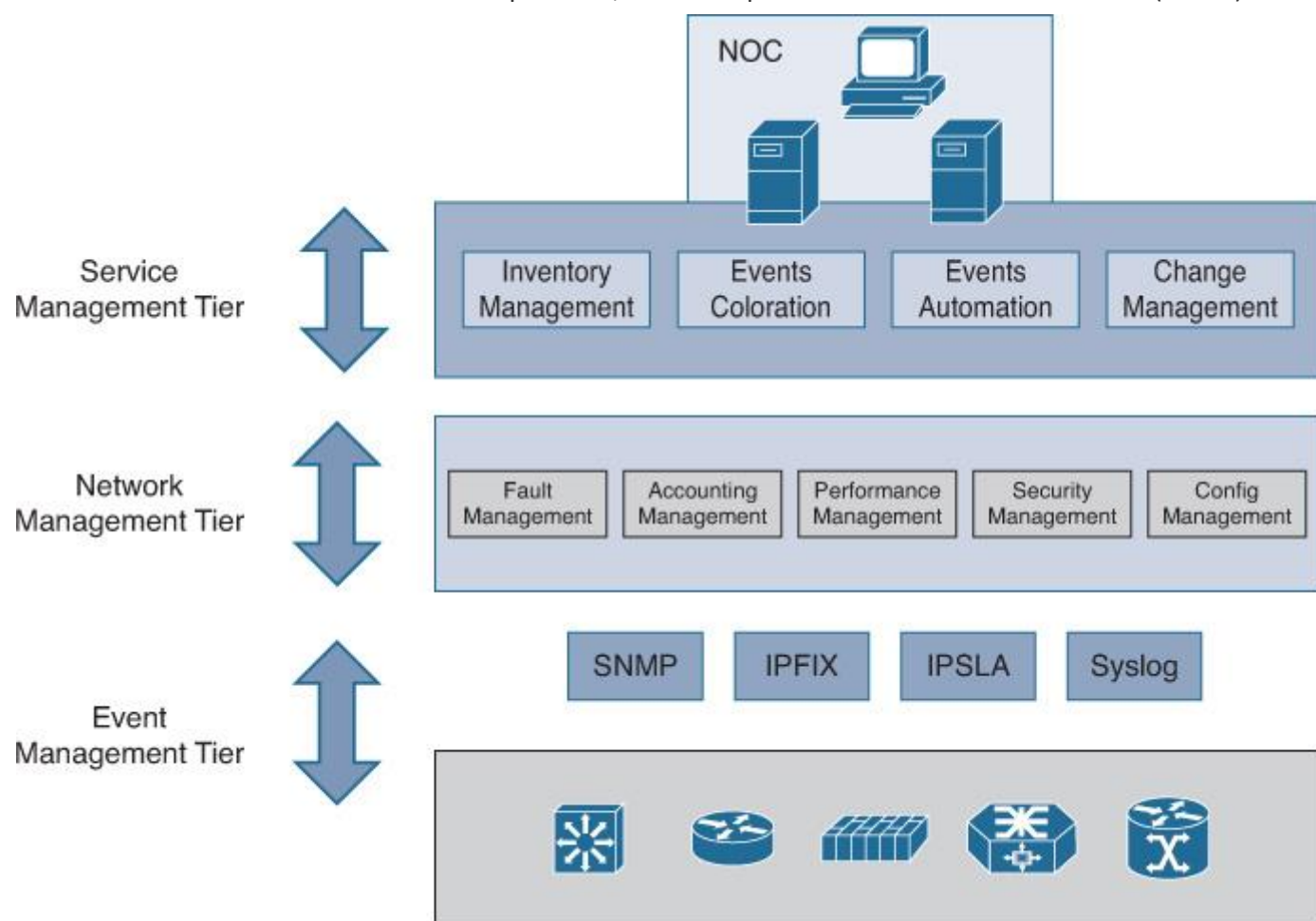


Figure 10-56 Multitiered Network Management Solution

This multitier network management approach is based on bottom-up communication flow between the different management systems and tools using various protocols, including NetFlow, syslog, and SNMP. In fact, in a large network, it is almost always impossible to cope with the number of events reported from each network element to the NMS at the higher layers. In addition, in certain situations, a failure or fault in specific areas within the network can impact multiple devices. Typically, each device will independently alert the NMS. As a result, there will be duplicate instances of the same problem in this case.

With this architecture, the network management tier (NMT) receives the input from multiple elements and applications, and then performs root-cause analysis by correlating the original information received from multiple sources and identifying the event that has occurred. This level of abstraction for event correlation provided by the NMT offers a simplified and efficient network management and operation solution. Network operators will be presented with the event deemed to be the most relevant and important, associated with the deduplication capabilities of network events, to reduce the number of unnecessary messages presented to the operations personnel. The service management tier in this approach provides an added intelligence and automation to the filtered events by the NMT and event correlation for more optimization, which will help network operators to move from complicated element management (per alert) to managing network events and identified problems [89]. The level of automation and intelligence provided by this approach helps network operators avoid the classic network management approach also known as *box by box*, where the network operator or administrator needs to visit every single network node and manually configure, manage, and monitor each device separately.

Further Reading

RFC 6632, *An Overview of the IETF Network Management Standards*, <http://www.ietf.org>

RFC 4377, *Operations and Management (OAM) Requirements for Multi-Protocol Label Switched (MPLS) Networks*, <http://www.ietf.org>

Information Technology Infrastructure Library (ITIL) Foundation, <https://en.wikipedia.org/wiki/ITIL>

Summary

This chapter covered various advanced IP topics and services that are part of any network design. To avoid design defects, network designers need to always incorporate these in an integrated holistic approach rather than designing in isolation. Moreover, considering the top-down design approach is a fundamental requirement to achieving a successful business-driven design (for example, ensuring that the design complies with the organization's security policy standards). This chapter also emphasized the importance of considering the business priorities and design constraints in which network design ideally must adopt the "first things first" approach, which takes into consideration existing limitations, which may include staff knowledge, budget, or supported features and technologies.

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